MILITARY SPECIFICATION

MICROCIRCUITS, LINEAR, SAMPLE AND HOLD CIRCUITS, MONOLITHIC SILICON

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for monolithic silicon, sample and hold circuits. Three product assurance classes and a choice of case outline/lead finish are provided for each type and are reflected in the complete part number.

1.2 Part number. The part number shall be in accordance with MIL-M-38510.

1.2.1 Device type. The device type shall be as follows:

Device type	Circuit
01	Sample and hold circuit, 10 kΩ load
02	Sample and hold circuit, 2 kΩ load

1.2.2 Device class. The device class shall be the product assurance level as defined in MIL-M-38510.

1.2.3 Case outline. The case outline shall be designated as follows:

Outline letter	Case outline (see MIL-M-38510, appendix C)
G	A-1 (8-lead can)
P	D-4 (8-lead, 1/4" x 3/8", dual-in-line package)

1.3 Absolute maximum ratings.

Supply voltage range	±18 V
Input voltage range 1/	±18 V
Logic to logic reference differential voltage	+7 V, -30 V
Storage temperature range	-65° to +150°C
Output short-circuit duration	Unlimited 2/
Lead temperature (soldering, 60 seconds)	300°C
Junction temperature (T_J)	175°C 3/
Hold capacitor short circuit duration	10 seconds

1.4 Recommended operating conditions.

Supply voltage range	±5 to ±15 V
Ambient operating temperature range	-55° to +125°C
Hold capacitor type	Teflon 4/

1/ The absolute maximum input voltage shall not exceed the power supply voltages.
2/ Short circuit may be to ground or either supply. Rating applies to +125°C case temperature or +75°C ambient temperature.
3/ For short term test (in the specific burn-in and life test configuration when required and up to 168 hours maximum), T_J = 275°C.
4/ Sample and hold performance is highly dependent on having a quality "hold" capacitor with low dielectric absorption over -55°C ≤ T_A ≤ 125°C.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Rome Air Development Center (RBE-2), Griffiss AFB, NY 13441, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:28Z
Check the source to verify that this is the current version before use.

1.5 Power and thermal characteristics.

Package	Case outline	Maximum allowable power dissipation 1/	Maximum θ_{J-C}	Maximum θ_{J-A}
8-lead can	G	330 mW @ T_A = +125°C	40°C/W	150°C/W
8-lead DIP	P	400 mW @ T_A = +125°C	35°C/W	120°C/W

1/ All leads welded or soldered to PC board.

2. APPLICABLE DOCUMENTS

2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-38510 - Microcircuits, General Specification for.

STANDARD

MILITARY

MIL-STD-883 - Test Methods and Procedures for Microelectronics.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-38510, and as specified herein.

3.2 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-M-38510 and herein.

3.2.1 Terminal connections and logic diagram. The terminal connections and logic diagram shall be as specified on figure 1.

3.2.2 Schematic circuits. The schematic circuits shall be as specified on figure 2.

3.2.3 Case outlines. The case outlines shall be as specified in 1.2.3.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-38510 (see 6.5).

3.4 Electrical performance characteristics. The electrical performance characteristics are specified in table I and, unless otherwise specified, apply over the full operating ambient temperature range for supply voltages from ±5 V to ±18 V. Unless otherwise specified, the device is in the "sample" mode with C_H = 0.01 µF. With ±15 V supplies, the input voltage range is from -11.5 V to +11.5 V. The logic reference voltage is 0 V and the logic voltage for "sample" mode is 2.5 V. Specific test conditions and limits are shown in table III.

3.4.1 Offset null circuit. The nulling inputs shall be capable of being nulled 1 millivolt beyond the specified offset voltage limits for -55°C $\leq T_A \leq$ 125°C using the circuit of figure 3.

3.4.2 Instability oscillations. The devices shall be free of oscillations when operated in the test circuits of this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:28Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical performance characteristics.

Test	Symbol	Conditions ($\pm V_{CC}$ = ±15 V, see figure 7 and 3.4, unless otherwise specified)		Limits Min	Limits Max	Unit
Input offset 1/ voltage	V_{IO}	$\pm V_{CC}$ = ±5 V to ±15 V	T_A = 25°C	-3	+3	mV
			-55°C ≤ T_A ≤ 125°C	-5	+5	mV
Input offset voltage temperature sensitivity	$\frac{\Delta V_{IO}}{\Delta T}$		T_A = -55°C to 25°C	-20	20	µV/°C
			T_A = 25°C to 125°C	-20	20	µV/°C
Input bias current 1/	I_{IB}	$\pm V_{CC}$ = ±5 V to ±15 V	T_A = 25°C	-1	25	nA
			-55°C ≤ T_A ≤ 125°C	-25	75	nA
Input impedance 2/	Z_I		T_A = 25°C	2	---	GΩ
			-55°C ≤ T_A ≤ 125°C	1	---	GΩ
Gain error	A_e	V_{IN} = -11.5 V to +11.5 V, R_L = 10 kΩ	T_A = 25°C	-.005	+.005	%
			-55°C ≤ T_A ≤ 125°C	-.02	+.02	%
		For device type 02 only V_{IN} = -10 V to +10 V, R_L = 2 kΩ	T_A = 25°C	-.005	+.005	%
			-55°C ≤ T_A ≤ 125°C	-.02	+.02	%
		V_{CC} = ±5 V, V_{IN} = -2 V to +2 V, R_L = 10 kΩ	T_A = 25°C	-.02	+.02	%
			-55°C ≤ T_A ≤ 125°C	-.04	+.04	%
Input offset voltage adjustment	$V_{IO(ADJ+)}$	V_{IN} = 0 V, $V_{OFFSET\ ADJ}$ at $+V_{CC}$		6	---	mV
	$V_{IO(ADJ-)}$	V_{IN} = 0 V, $V_{OFFSET\ ADJ}$ at 1 kΩ from $+V_{CC}$, 20 kΩ from 0 V		---	-6	mV
Power supply rejection ratio	+PSRR	$+V_{CC}$ = +12 V to +18 V, $-V_{CC}$ = -18 V, V_{IN} = 0 V		80	---	dB
	-PSRR	$+V_{CC}$ = +18 V, $-V_{CC}$ = -12 V to -18 V, V_{IN} = 0 V		80	---	dB
Feedthrough rejection ratio 3/	FRR	V_{IN} = ±11.5 V, Hold mode (see fig. 8)	T_A = 25°C	86	---	dB
			-55°C ≤ T_A ≤ 125°C	80	---	dB
	FRR_{ac}	V_{IN} = 20 V_{pp} @ 1 kHz, Hold mode (see fig. 9)	T_A = 25°C	86	---	dB
Series change resistance 4/	R_{SC}	V_{IN} = 0 V to 0.4 V; measure current change to ground at pin 6 (hold cap. terminal)		100	400	Ω
Output impedance	Z_O	Hold mode (see fig. 10) $V_{hold\ cap.}$ = V_{HC} = 0 V, I_O = ±1 mA		---	2	Ω

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:28Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions ($\pm V_{CC}$ = ±15 V, see figure 7 and 3.4, unless otherwise specified)		Limits Min	Limits Max	Unit
"Hold" step voltage 5/	V_{HS}	V_{Logic} = 4 V (see fig. 11) $t_r \leq 50$ ns, V_{OUT} = ±11.5 V	T_A = 25°C	-2	2	mV
			-55°C ≤ T_A ≤ 125°C	-5	5	mV
Supply current	I_{CC}	T_A = -55°C	Device type 01	1	6.5	mA
			Device type 02	1	7.0	mA
		25°C ≤ T_A ≤ 125°C	Device type 01	1	5.5	mA
			Device type 02	1	6.0	mA
Logic input current (high)	I_{IH}	V_{Logic} = 5.5 V, $+V_{CC}$ = 8.5 V, $-V_{CC}$ = -21.5 V		0	10	µA
Logic input current (low)	I_{IL}	V_{Logic} = 0 V, $+V_{CC}$ = 21.5 V, $-V_{CC}$ = -8.5 V		-1	1	µA
Output short circuit current (positive output)	$I_{OS(+)}$	V_{IN} = 10 V, t ≤ 25 ms (short circuit to 0 V)	Device type 01	-25	---	mA
			Device type 02	-30	---	mA
Output short circuit current (negative output)	$I_{OS(-)}$	V_{IN} = -10 V, t ≤ 25 ms (short circuit to 0 V)	Device type 01	---	25	mA
			Device type 02	---	30	mA
Hold mode leakage current (positive output)	$I_{HL(+)}$ 6/	V_{IN} = 0 V, $+V_{CC}$ = 3.5 V, $-V_{CC}$ = -26.5 V (see fig. 12)	T_J = 25°C 7/	-100	100	pA
			T_J = 125°C 7/	-50	50	nA
Hold mode leakage current (negative output)	$I_{HL(-)}$ 6/	V_{IN} = 0 V, $+V_{CC}$ = 26.5 V, $-V_{CC}$ = -3.5 V (see fig. 12)	T_J = 25°C 7/	-100	100	pA
			T_J = 125°C 7/	-50	50	nA
Hold capacitor charge current (positive output)	$I_{CH(+)}$	V_{IN} = 11.5 V, V_{HC} = 9.5 V	T_A = 25°C	---	-3	mA
			-55°C ≤ T_A ≤ 125°C	---	-2	mA
Hold capacitor charge current (negative output)	$I_{CH(-)}$	V_{IN} = -11.5 V, V_{HC} = -9.5 V	T_A = 25°C	3	---	mA
			-55°C ≤ T_A ≤ 125°C	2	---	mA
Differential logic threshold	$V_{TH(H)}$	V_{IN} = -2 V, V_{Logic} = 2 V (check for I_{hold} cap. ≥ 1 mA)		---	2	V
	$V_{TH(L)}$	V_{IN} = -2 V, V_{Logic} = 0.8 V (check for I_{hold} cap. ≤ 10 µA)		.8	---	V
Acquisition time (0.01% error)	t_{aq} 8/	V_{IN} = 0 V to 10 V, 10 V to 0 V, 0 V to -10 V, -10 V to 0 V, C_L = 100 pF (see fig. 13 thru 15)	T_A = 25°C	---	25	µs

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:28Z
Check the source to verify that this is the current version before use.

TABLE I. Electrical performance characteristics - Continued.

Test	Symbol	Conditions ($\pm V_{CC}$ = ±15 V, see figure 7 and 3.4, unless otherwise specified)		Limits Min	Limits Max	Unit
Aperture time	t_{ap} 9/	V_{IN} = 0 V to +10 V, 10 V to 0 V, 0 V to -10 V, -10 V to 0 V, $V_O \leq$ 1 mV (see fig. 16 thru 19)	T_A = 25°C	---	300	ns
Transient response (settling time) 10/	$TR_{(ts)}$	V_{IN} = 100 mV step, C_H = 1000 pF, R_L = 10 kΩ, C_L = 100 pF (see fig. 20) (to 10% of final value)	T_A = 25°C	---	2.5	µs
Transient response (overshoot) 10/	$TR_{(os)}$	V_{IN} = 100 mV step, C_H = 1000 pF, R_L = 10 kΩ, C_L = 100 pF (see fig. 20)	T_A = 25°C	---	40	%
Noise	$e_{n(H)}$ $e_{n(S)}$	Hold mode, sample mode 10 Hz to 10 kHz (see fig. 21)	T_A = 25°C	---	10	µVrms
Settling time	ts	V_{IN} = 0 V, $V_O \leq$ 1 mV, hold mode (see fig. 22)	T_A = 25°C	---	1.5	µs

1/ This parameter is specified at V_{CM} = 0 V, -11.5 V, and +11.5 V with $\pm V_{CC}$ = ±15 V, and at V_{CM} = -2 V and +2 V with $\pm V_{CC}$ = ±5 V.

2/ Input impedance is calculated from the V_{IO} and I_{IB} common mode voltage end-point range data.

3/ Feedthrough rejection ratio is very sensitive to stray capacitance between the signal input (pin 3) and the hold capacitor (pin 6). For instance 0.5 pF of external coupling with a .01 µF hold capacitor would equal the specification limit of the device.

(i.e., $FRR = 20 \log \frac{(.01\ \mu F)}{.5\ pF} = 20 \log (2.10^4) = 86$ dB).

4/ Series change resistance along with input signal slew rate and an external hold capacitor determine the dynamic sampling error of the device in its application (i.e., $DSE = K \cdot R_{SC} \cdot SR$ where K is a proportionality constant).

5/ The external hold capacitor should be either Teflon or polystyrene so that dielectric absorption is minimized. This will insure that excessive "sag back" after capacitor "sample" mode charging does not occur. "Hold" step is sensitive to stray capacitive coupling between input logic signals and the "hold" capacitor.

6/ Hold mode leakage current is actually JFET junction leakage current which doubles (approximately) for each 10°C increase in junction temperature. Measurement at -55°C is not necessary since expected values are too small for typical test systems.

7/ The specification is at T_J instead of T_A to enable automatic testing. A warm-up time delay would be required for T_A testing.

8/ Acquisition time at 125°C typically increases from 20% to 100% above the 25°C value.

9/ Aperture time at 125°C typically increases 100% above the 25°C value.

10/ Transient response shall be measured at the common mode voltage limits (i.e., V_{CM} = -11.5 V and +11.5 V). Any high frequency ringing shall be over within 1 microsecond. After its peak the major loop response shall be without further oscillations.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:28Z
Check the source to verify that this is the current version before use.

3.5 Electrical test requirements. The electrical test requirements shall be as specified in table II. The subgroups of table III which constitute the minimum electrical test requirements for screening, qualification, and quality conformance by device class are specified in table II.

3.6 Marking. Marking shall be in accordance with MIL-M-38510. At the option of the manufacturer, marking the country of origin may be omitted from the body of the microcircuit, but shall be retained on the initial container.

3.7 Microcircuit group assignment. The devices covered by this specification shall be in microcircuit group number 60 (see MIL-M-38510, appendix E).

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-M-38510 and methods 5005 and 5007, as applicable, of MIL-STD-883, except as modified herein.

4.2 Screening. Screening shall be in accordance with method 5004 of MIL-STD-883, and shall be conducted on all devices prior to qualification and quality conformance inspection. The following additional criteria shall apply:

a. Burn-in test (method 1015 of MIL-STD-883).

(1) Class S devices - Test condition D using the circuit shown on figure 4.

(2) Class B devices - Test condition D using the circuit shown on figure 4, or test condition C using the circuit shown on figure 5, or test condition F using the circuit shown on figure 6.

NOTE: If accelerated high-temperature test conditions are used, the device manufacturer shall ensure that at least 85 percent of the applied voltage is dropped across the device at temperature. The device is not considered functional under accelerated test conditions.

b. Reverse bias burn-in (method 1015 of MIL-STD-883). This screen test shall apply to class S devices only using the circuit shown on figure 5.

c. Interim and final electrical test parameters shall be as specified in table II herein, except interim electrical parameters test prior to burn-in is optional at the discretion of the manufacturer.

d. The percent defective allowable (PDA) for class S devices shall be as specified in MIL-M-38510. The PDA for class B devices shall be 10 percent based on failures from group A, subgroup 1 test after cooldown as final electrical test in accordance with method 5004 of MIL-STD-883, and with no intervening electrical measurements. If interim electrical parameter tests are performed prior to burn-in, failures resulting from pre burn-in screening may be excluded from the PDA. If interim electrical parameter tests prior to burn-in are omitted, then all screening failures shall be included in the PDA. The verified failures of group A, subgroup 1 after burn-in divided by the total number of devices submitted for burn-in in that lot shall be used to determine the percent defective for that lot, and the lot shall be accepted or rejected based on the PDA for the applicable device class.

4.3 Qualification inspection. Qualification inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-M-38510. Inspections to be performed shall be those specified in method 5005 of MIL-STD-883 and herein for groups A, B, C, and D inspections (see 4.4.1 through 4.4.4).

Source: https://assist.dla.mil -- Downloaded: 202610-21T13:28Z
Check the source to verify that this is the current version before use.

a. Component values are nominal.

TABLE II. Electrical test requirements.

MIL-STD-883 test requirements	Subgroups (see table III)		
	Class S devices	Class B devices	Class C devices
Interim electrical parameters (pre burn-in) (method 5004)	1	1	None
Final electrical test parameters (method 5004)	1*,2,3	1*,2,3	1
Group A test requirements (method 5005)	1,2,3,7,8	1,2,3,7,8	1,2,3,7,8
Group C end-point and group B, class S, electrical parameters (method 5005)	1,2,3 and table IV delta limits	1 and table IV delta limits	1 and table IV delta limits
Group D end-point electrical parameters (method 5005)	1,2,3	1	1

*PDA applies to subgroup 1 (see 4.2d).

4.4.1 Group A inspection. Group A inspection shall be in accordance with table I of method 5005 of MIL-STD-883 and as follows:

a. Tests shall be as specified in table II herein.

b. Subgroups 4, 5, 6, 9, 10, and 11 shall be omitted.

4.4.2 Group B inspection. Group B inspection shall be in accordance with table II of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein.

b. Life test for class S devices shall be in accordance with table IIa (subgroup 5) of method 5005 of MIL-STD-883, using the circuit on figure 6. If the alternate burn-in conditions are used, the circuit on figure 5 shall be used.

4.4.3 Group C inspection. Group C inspection shall be in accordance with table III of method 5005 of MIL-STD-883 and as follows:

a. End-point electrical parameters shall be as specified in table II herein. Delta limits shall apply only to subgroup 1 of group C inspection for class B and C devices.

b. Life test for class B and C devices (method 1005 of MIL-STD-883). Test condition D using the circuit shown on figure 5, or test condition F using the circuit shown on figure 6 (see note following 4.2a.(2)).

4.4.4 Group D inspection. Group D inspection shall be in accordance with table IV of method 5005 of MIL-STD-883. End-point electrical parameters shall be as specified in table II herein.

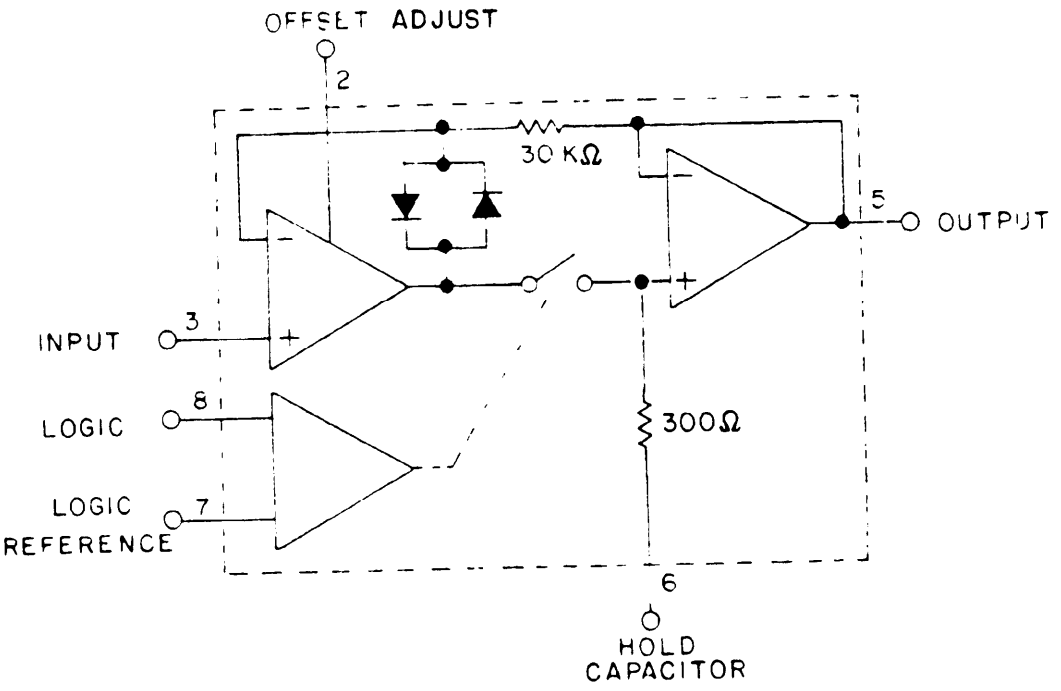
4.4.5 Inspection of packaging. The sampling and inspection of the preservation, packing, and container marking shall be in accordance with the requirements of MIL-M-38510.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows:

4.5.1 Voltage and current. All voltages given, except the input offset voltage (or differential voltage), are referenced to the external zero reference level of the supply voltage. Currents given are conventional and positive when flowing into the referenced terminal.

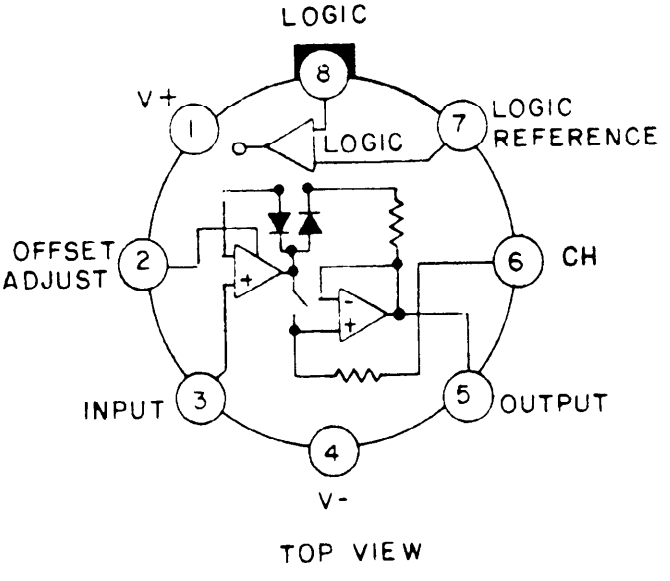
Text continues on page 36.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:28Z
Check the source to verify that this is the current version before use.



Logic Diagram

Device types 01, 02
(case G)



TOP VIEW

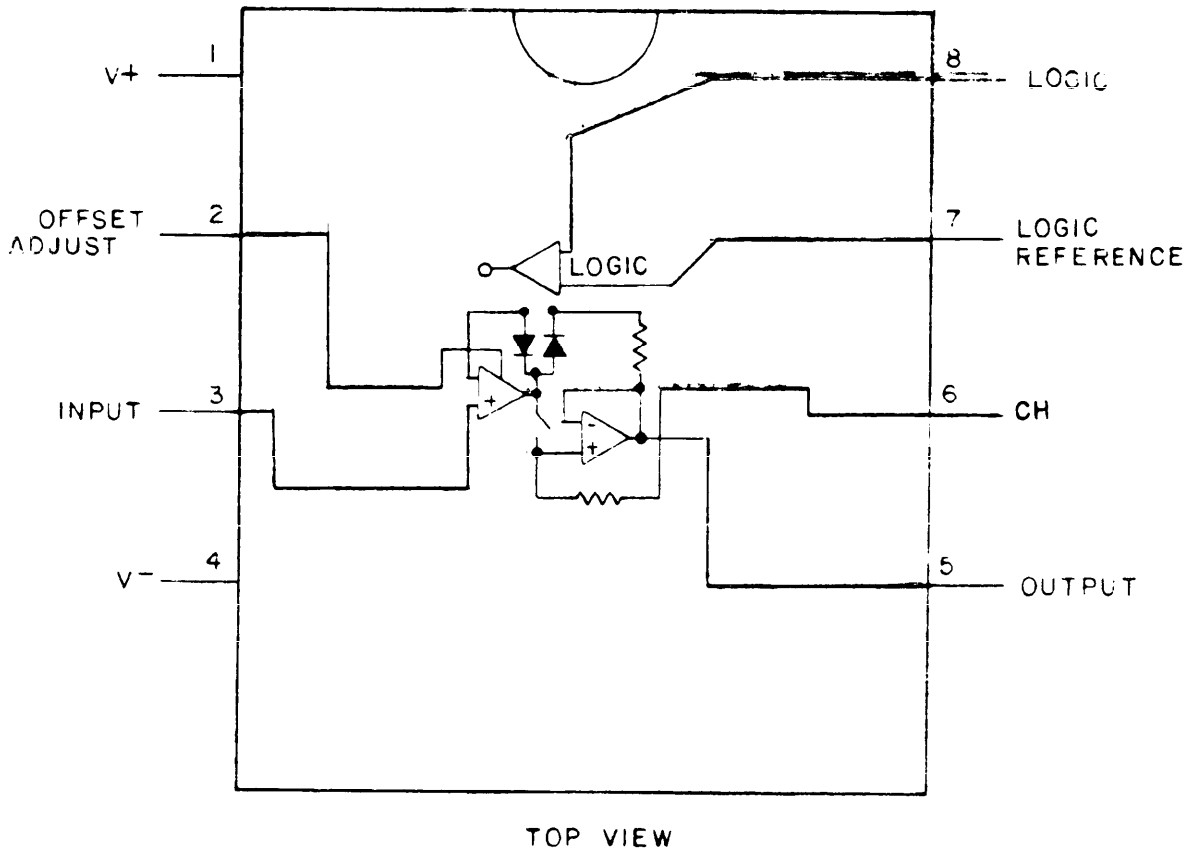
(Pin 4 connected to case)
8 lead can (top view)

FIGURE 1. Terminal connections and logic diagram.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:28Z
Check the source to verify that this is the current version before use.

Device type 01.

(Case P)



8 lead dual-in-line (top view)

FIGURE 1. Terminal connections and logic diagram - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:28Z
Check the source to verify that this is the current version before use.

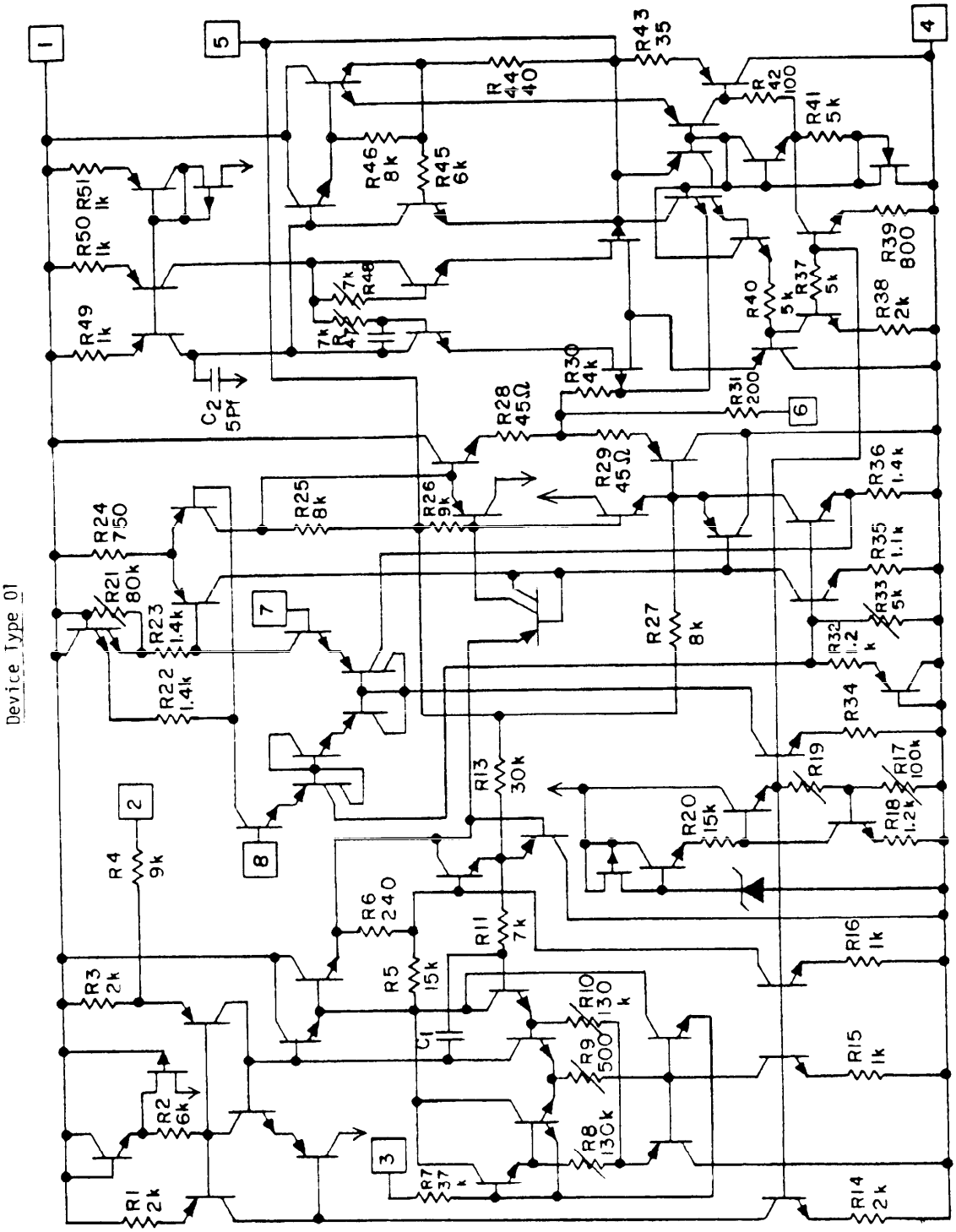
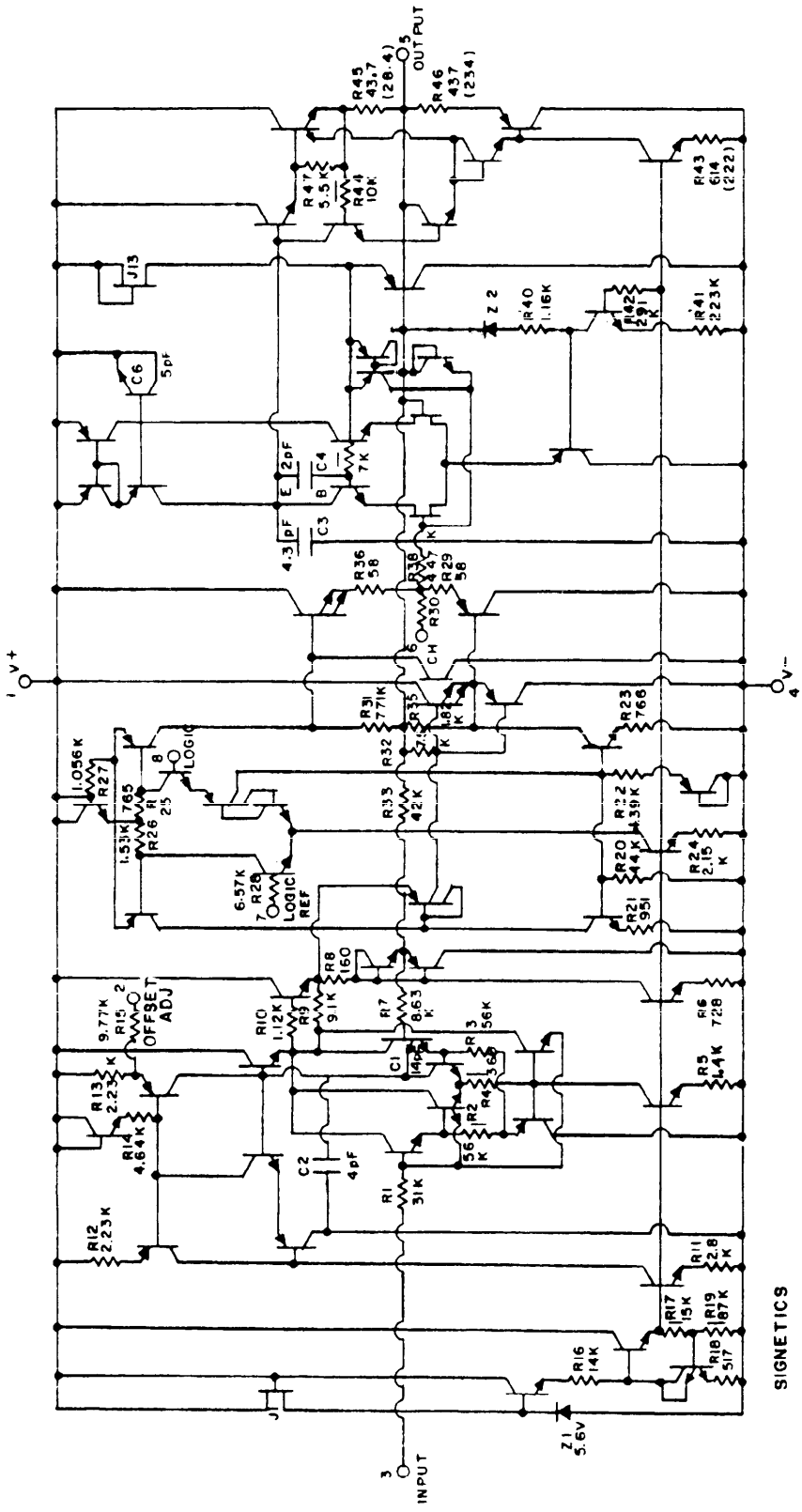


FIGURE 2. Schematic circuits.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:28Z
Check the source to verify that this is the current version before use.

Device types 01, 02



NOTE: All resistance and capacitance values are nominal.

FIGURE 2. Schematic circuits - Continued.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:28Z
Check the source to verify that this is the current version before use.

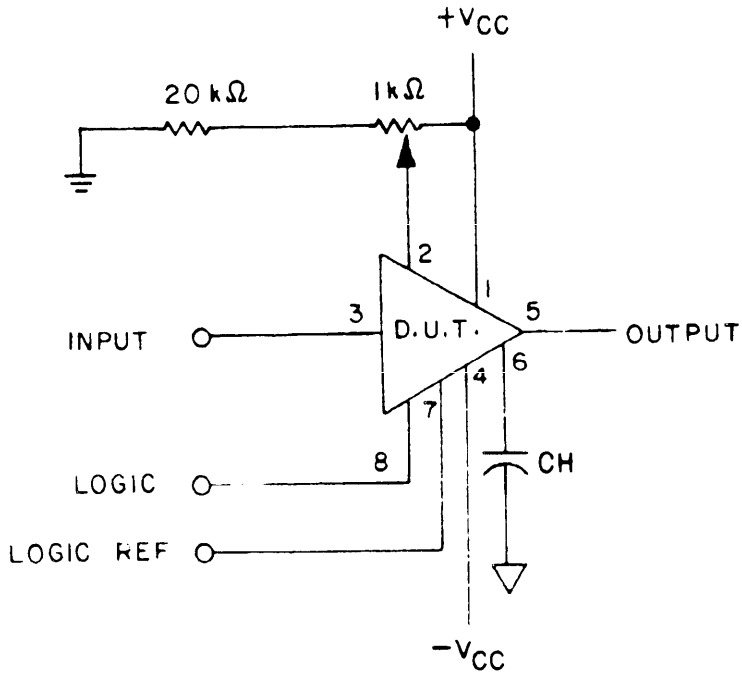
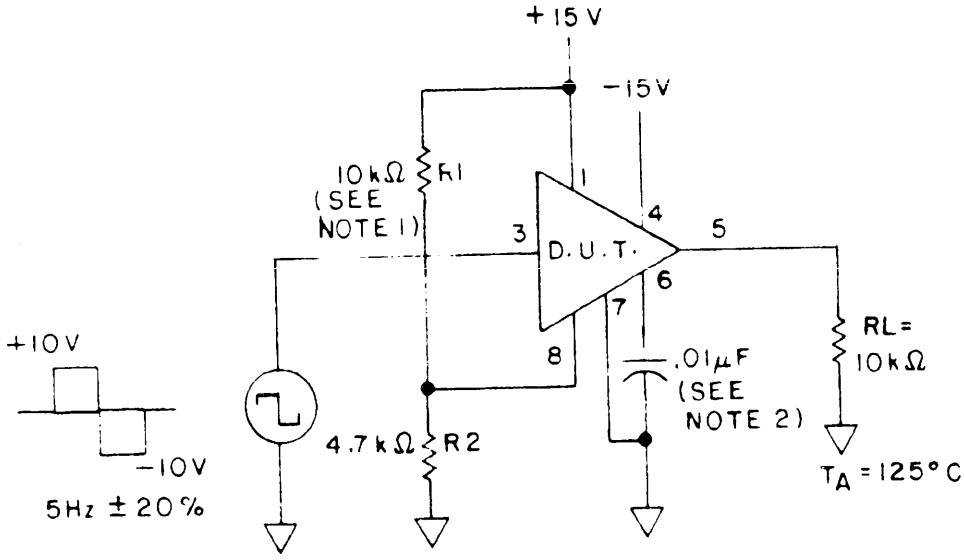


FIGURE 3. - Offset null circuit



NOTES:

1. R1 and R2 may be shared with many devices.
2. A quality S/H grade capacitor is not required for this application.

FIGURE 4. - Test circuit for burn-in and operating life tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:28Z
Check the source to verify that this is the current version before use.

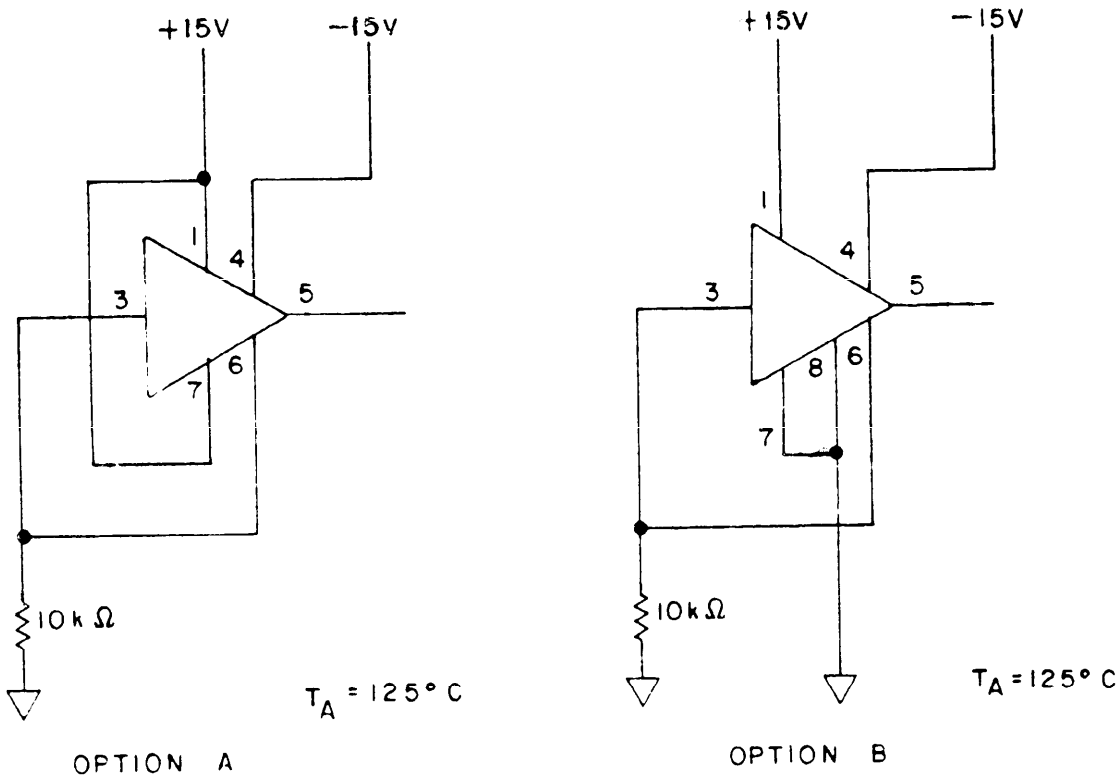


FIGURE 5. - Test circuit, burn-in (steady state power and reverse bias) and operating life test

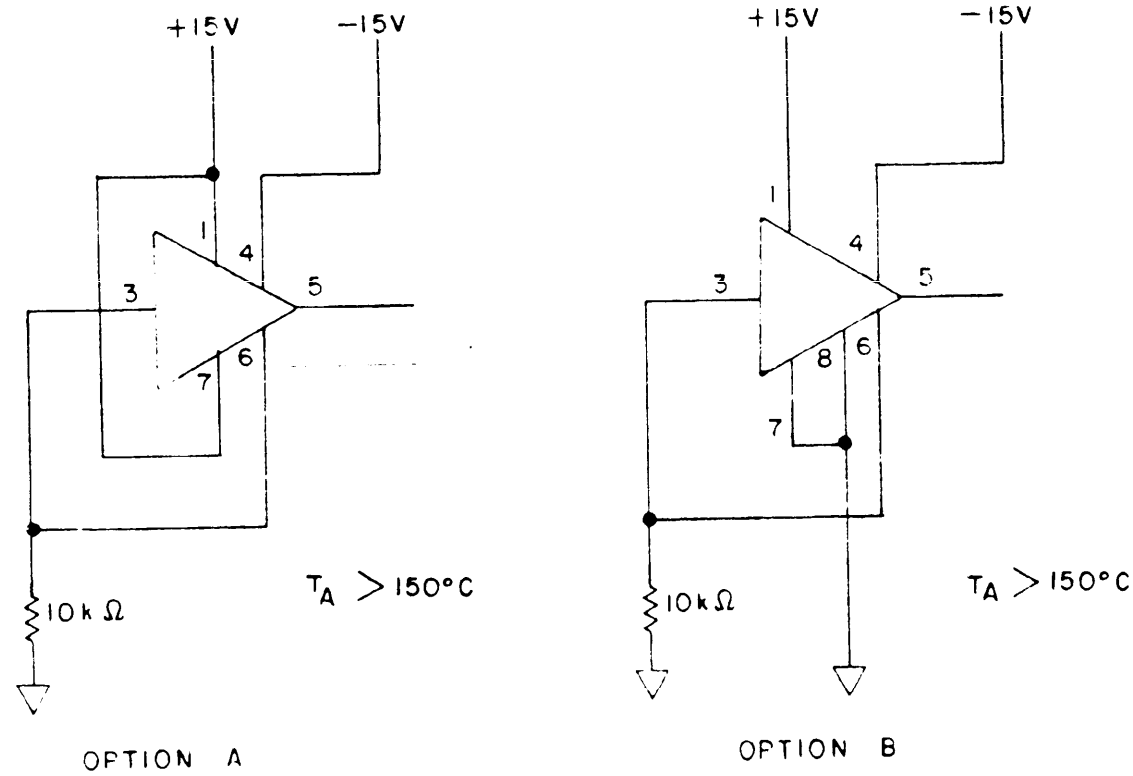
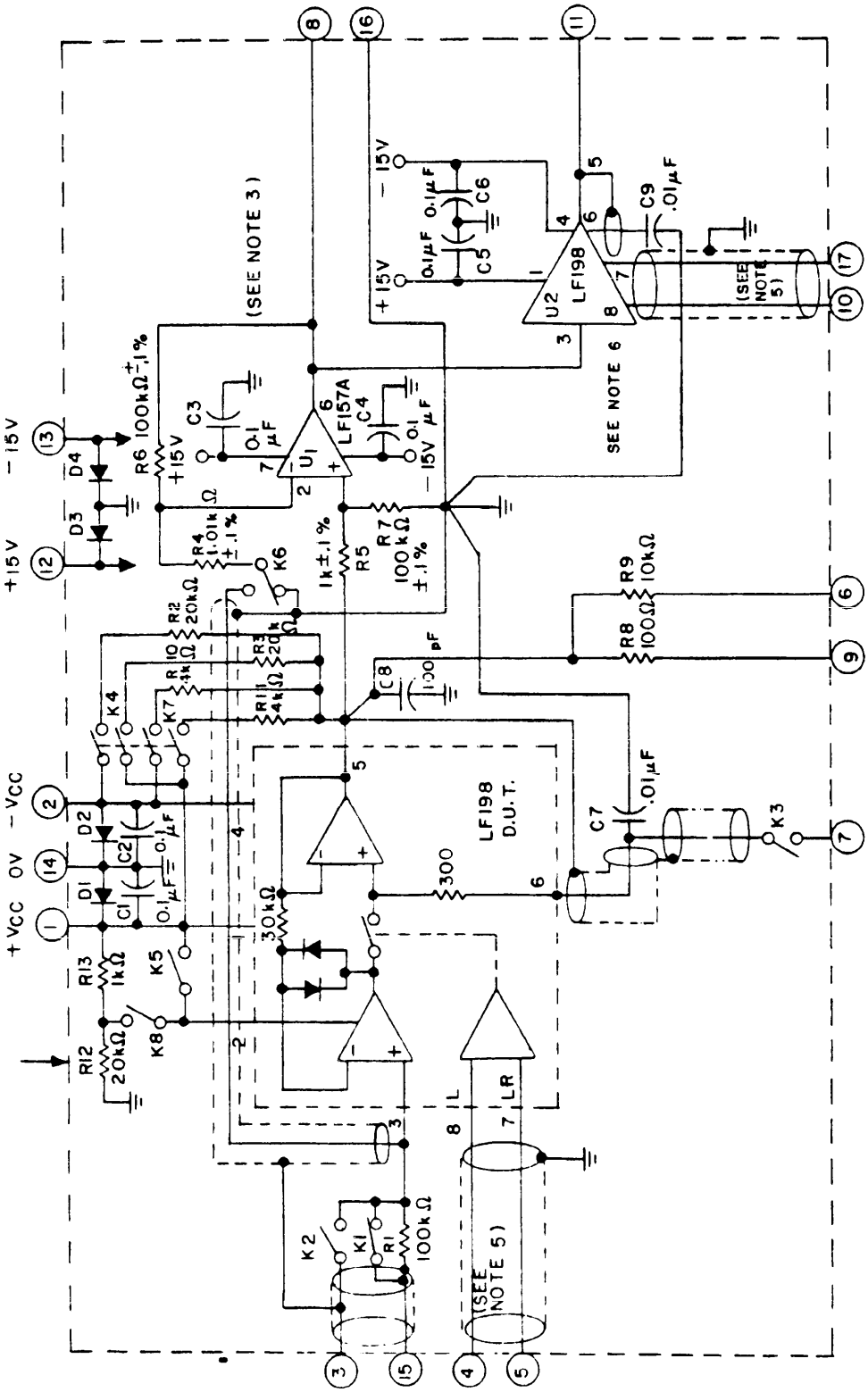


FIGURE 6. - Accelerated burn-in and life test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:28Z
Check the source to verify that this is the current version before use.

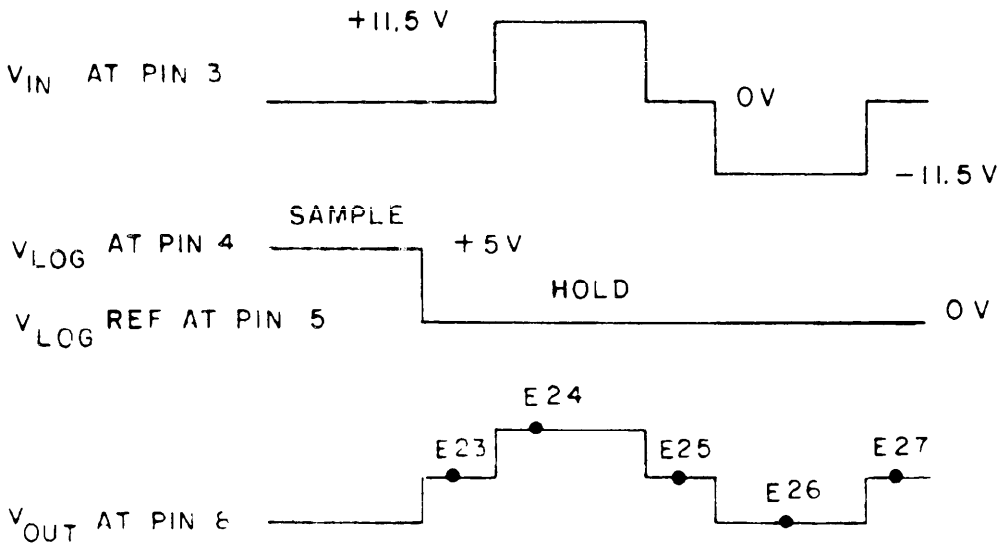


NOTES:

1. Last component designations are R13, C9, U2, D4 and K8.
2. All resistors are ± 1% film unless otherwise stated.
3. U1 offset error shall be measured with Pin 9 grounded and the D.U.T. removed. This error shall be removed in the software calculations.
4. Relay control inputs are not shown.
5. A speed-up circuit is required in series with the logic input for rise times greater than 0.5 µs.
6. The adapter S/H U2 is required for T_A = 125°C testing of some parameters.

FIGURE 7. Test circuit for static and dynamic tests.

Source: https://assist.dla.mil -- Downloaded: 202610-21T13:28Z
Check the source to verify that this is the current version before use.



FRR = 20 log 1150/E24 - E23

FRR = 20 log 1150/E25 - E24

FRR = 20 log 1150/E26 - E25

FRR = 20 log 1150/E27 - E26

FIGURE 8. Feedthrough rejection test timing waveforms.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:28Z
Check the source to verify that this is the current version before use.

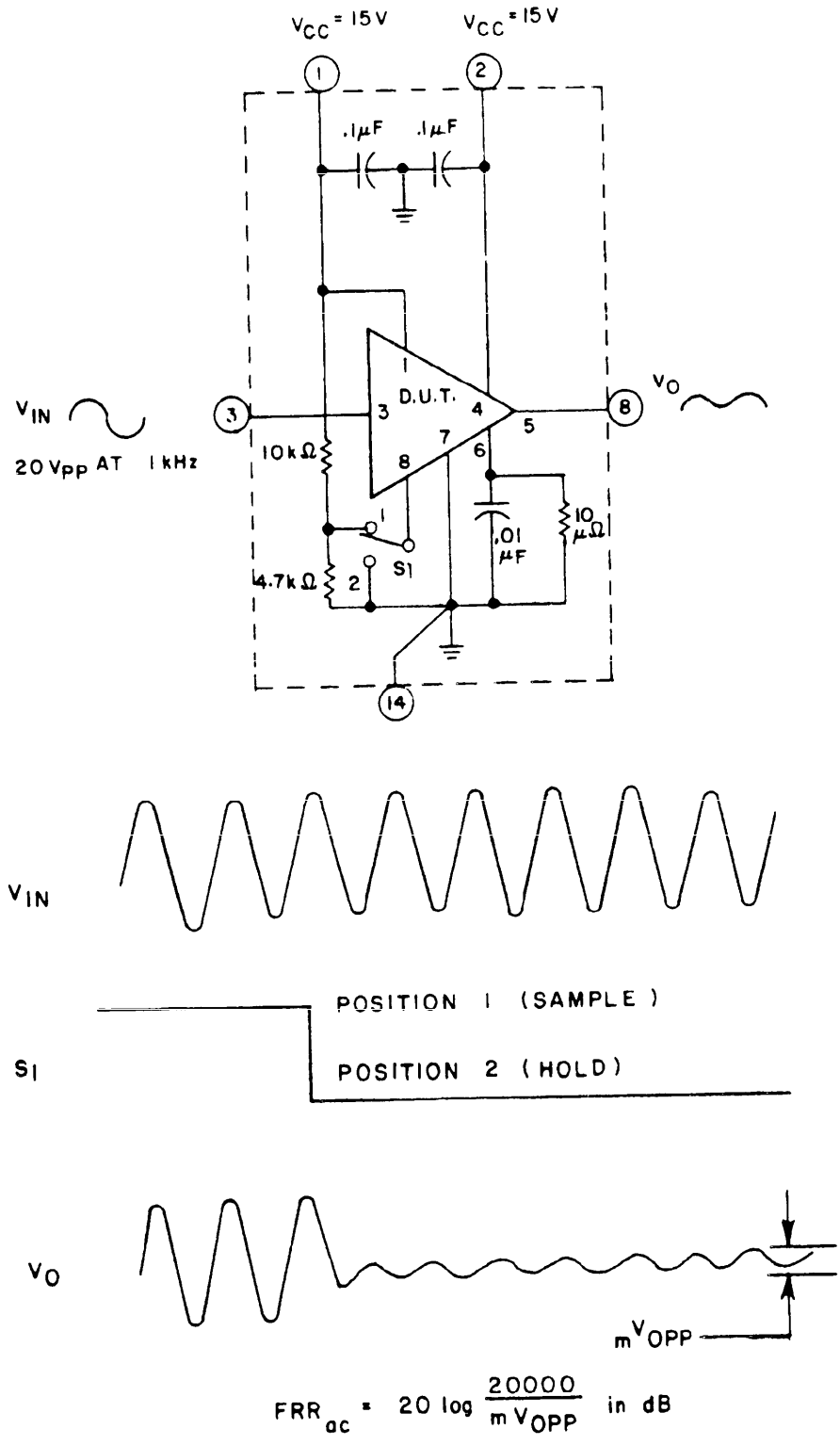
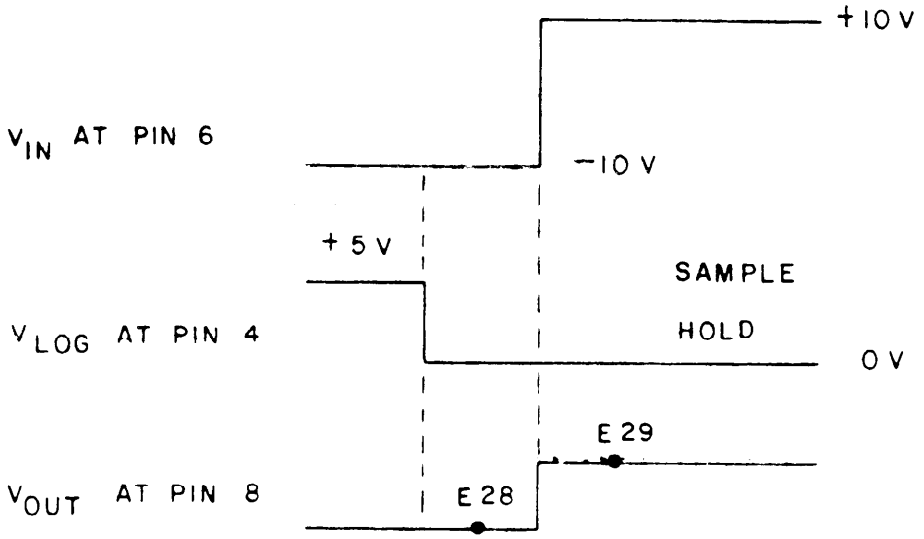


FIGURE 9. Test circuit for ac feedthrough rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:28Z
Check the source to verify that this is the current version before use.



V_{HC} AT PIN 7 = 0V (K3 IS ENERGIZED)

$Z_o = 5 (E29 - E28)$

FIGURE 10. - Output Impedance Test Timing

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:28Z
Check the source to verify that this is the current version before use.

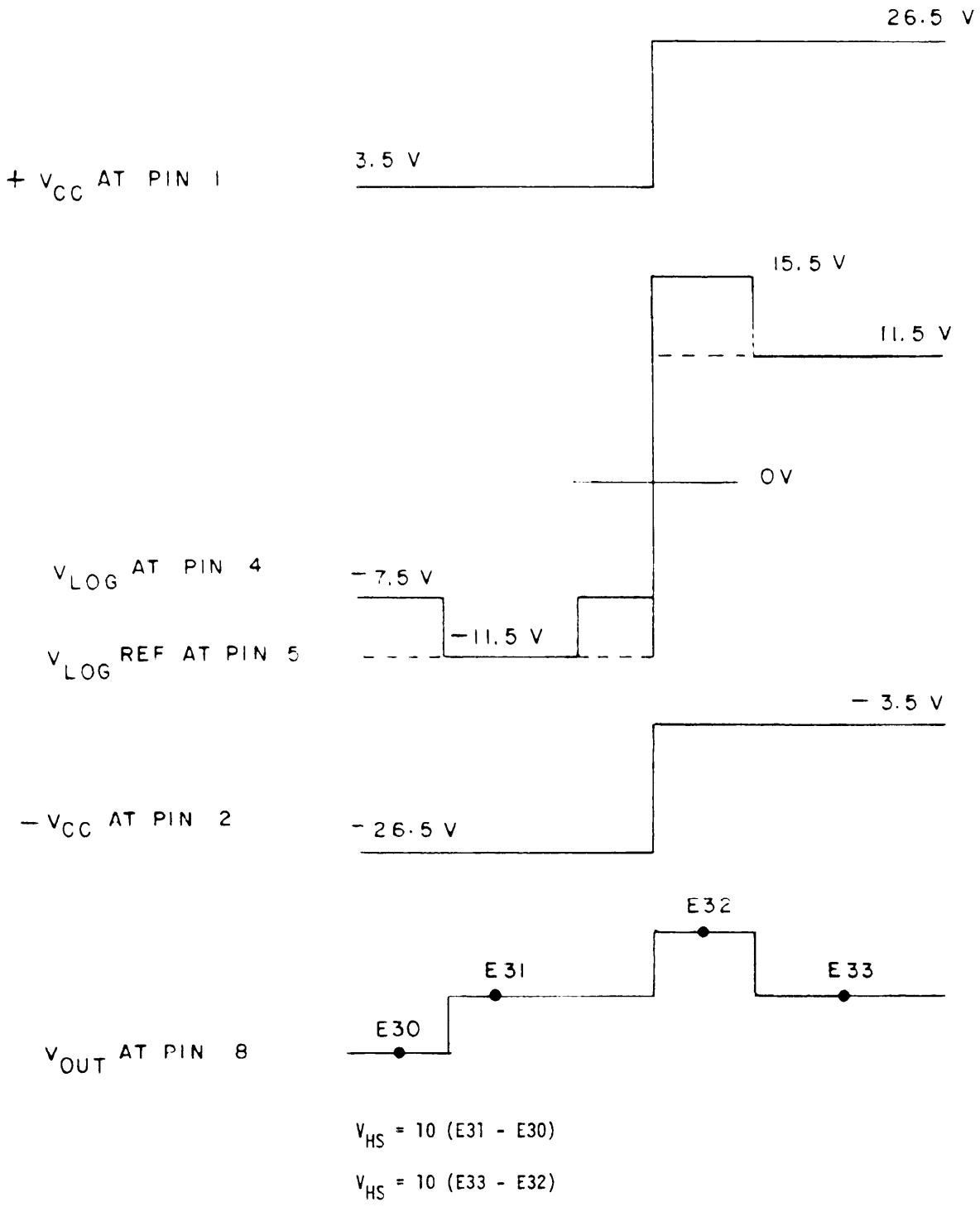
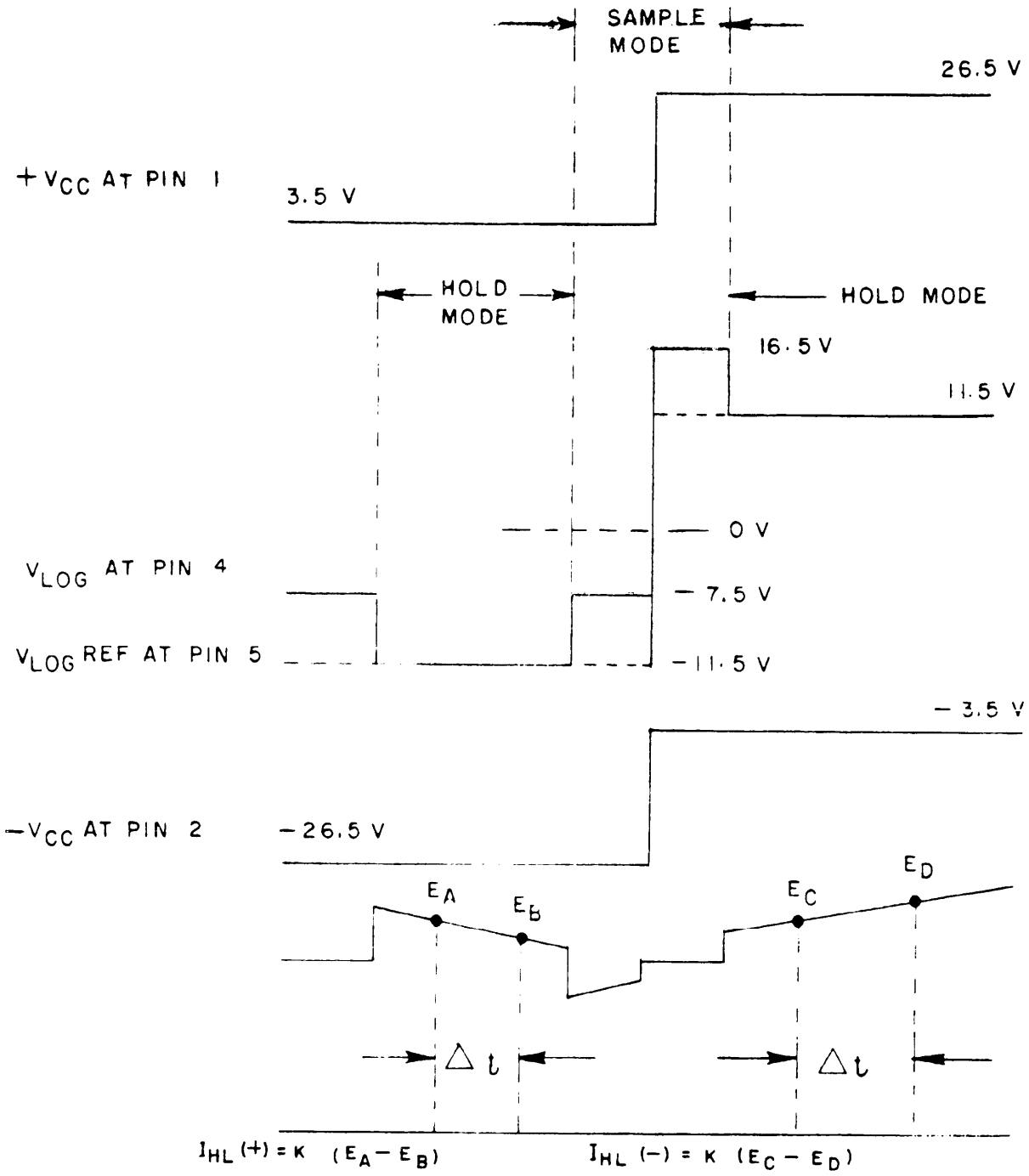


FIGURE 11. - Hold Step Timing Waveforms

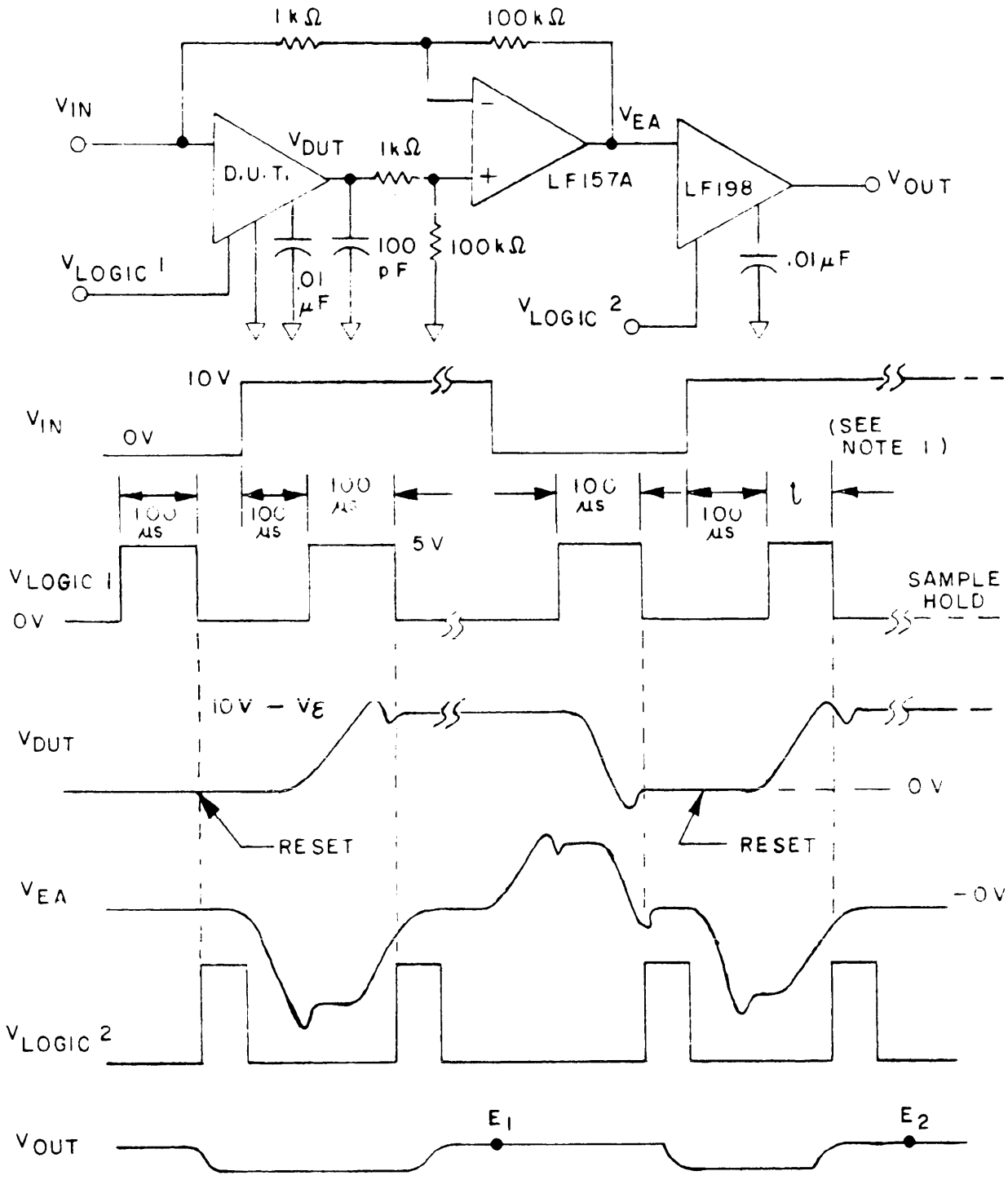
Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:28Z
Check the source to verify that this is the current version before use.



Temp	Δ t	K	ΔV_0 Units	I_L Units
25°C	100 ms	1	mV	pA
125°C	10 ms	0.1	mV	nA

FIGURE 12. - Hold Leakage Current Timing Waveforms

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:28Z
Check the source to verify that this is the current version before use.



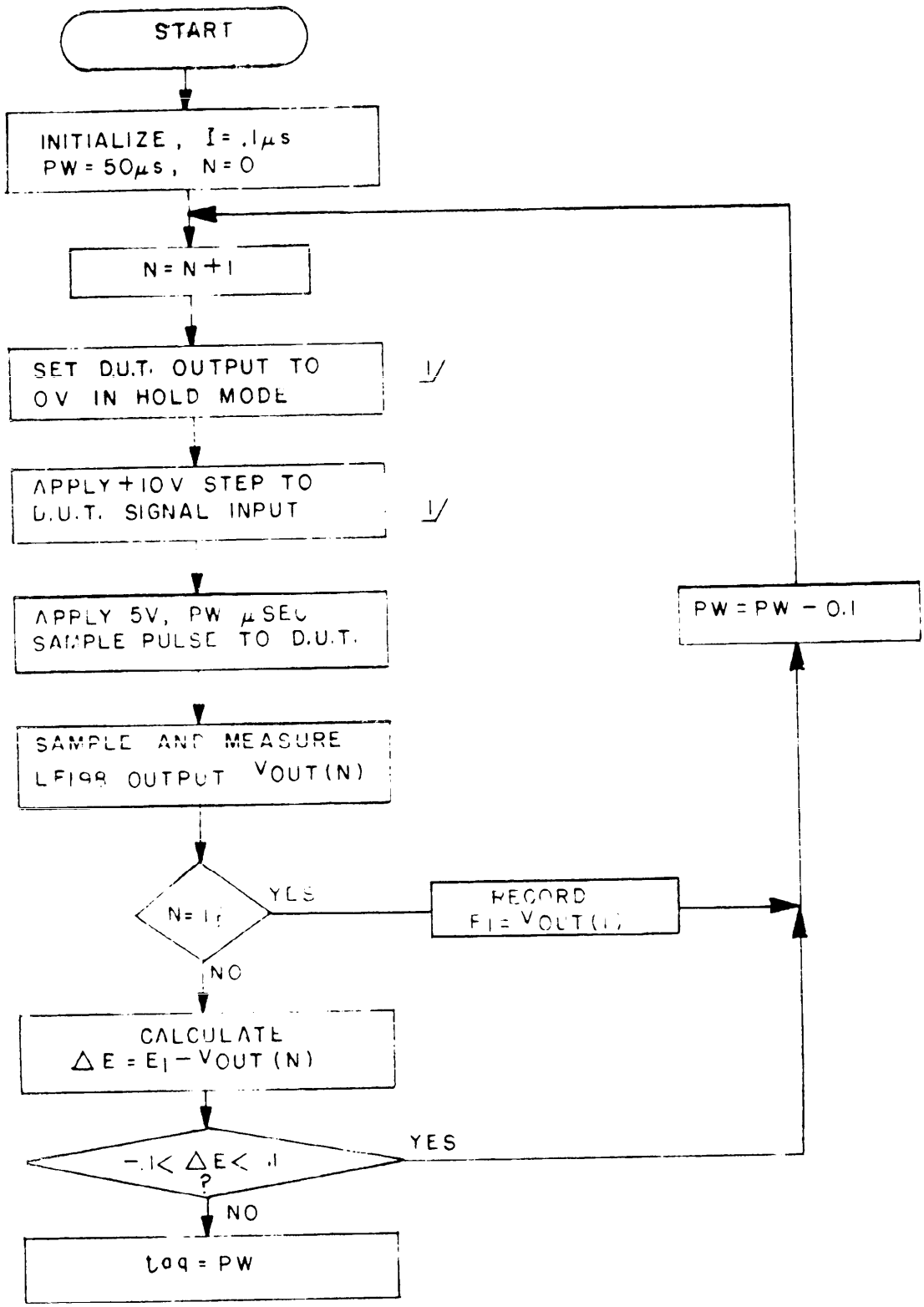
NOTES:

1/ Acquisition time is equal to the logic sample pulse width which yields an output error from steady state of 1 mV at the DUT output or .1 V at the diff. amp output (i.e., pulse width t = taq when $E_1 - E_2$ = .1 V).

2/ Only the 0 V to + 10 V step is shown. The other transitions as per Figure 14 shall also be tested.

FIGURE 13. Acquisition time test circuit and waveforms.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:28Z
Check the source to verify that this is the current version before use.



1/ Repeat above procedure for three other cases as follows: - 10 V step from 0, - 10 V step from + 10 V and + 10 V step from - 10 V

FIGURE 14. Acquisition time flow chart.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:28Z
Check the source to verify that this is the current version before use.

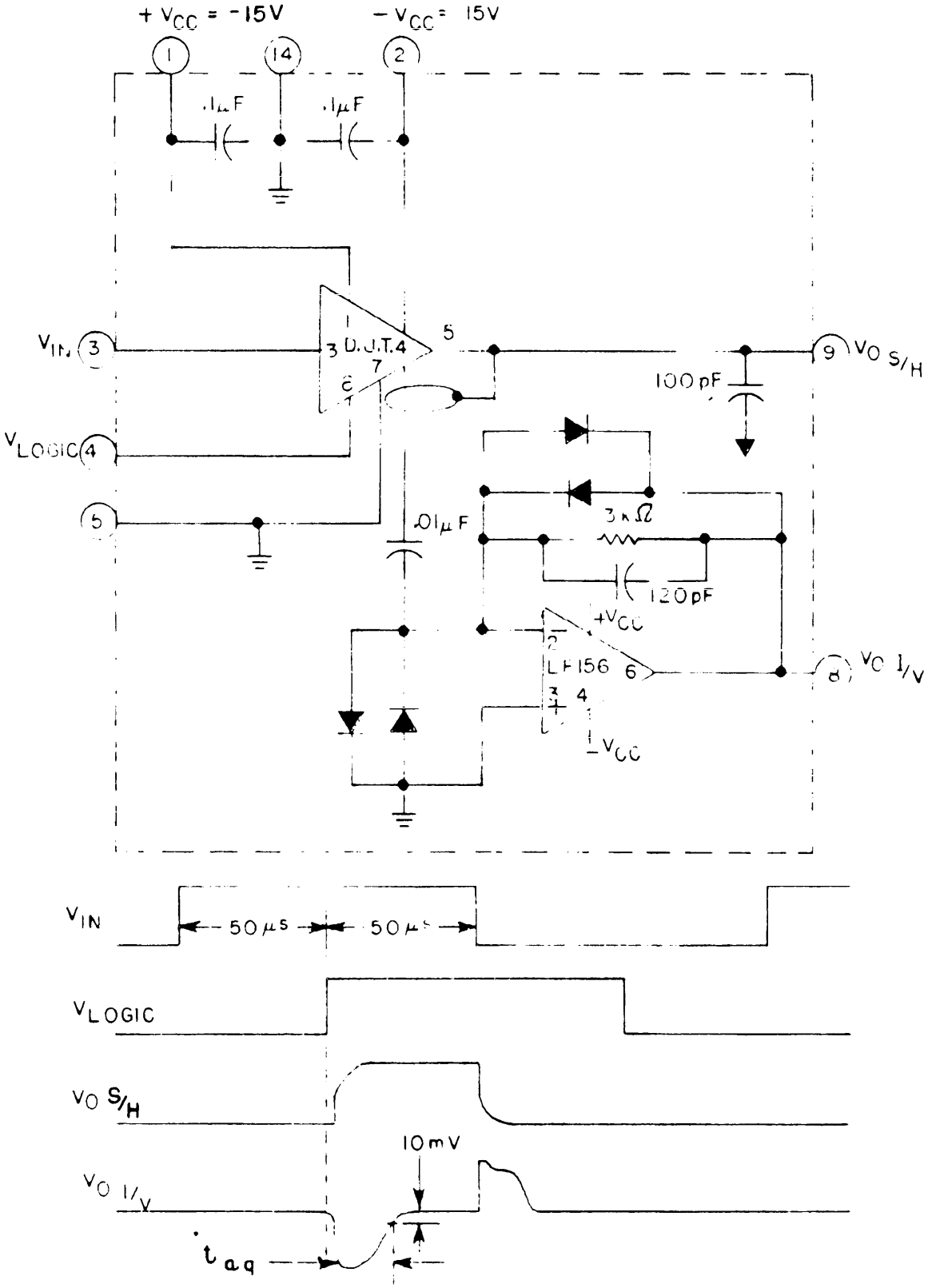
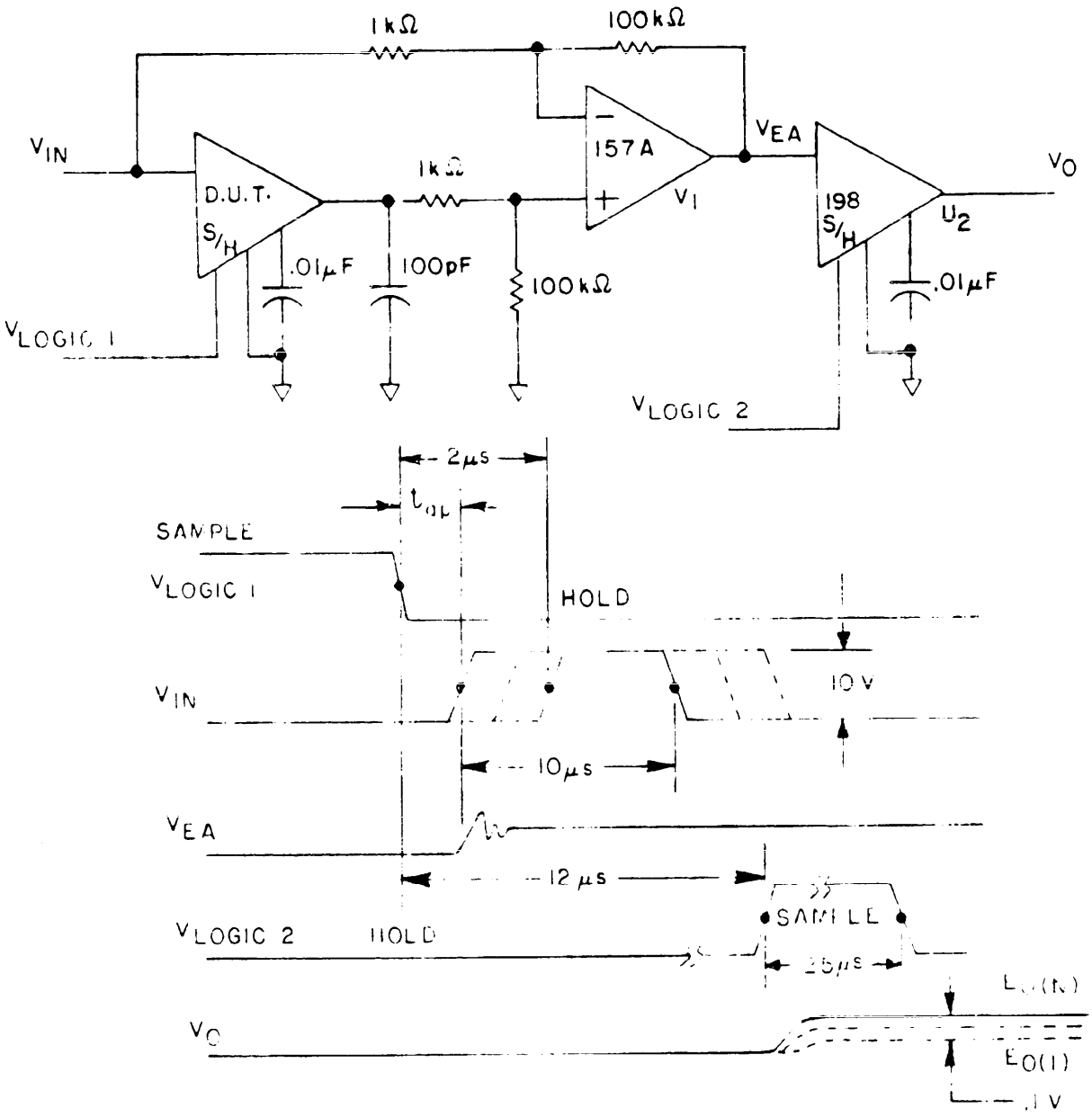


FIGURE 15. Alternate acquisition time test circuit.

Source: https://assist.dla.mil -- Downloaded: 202610-21T13:28Z
Check the source to verify that this is the current version before use.



NOTES:

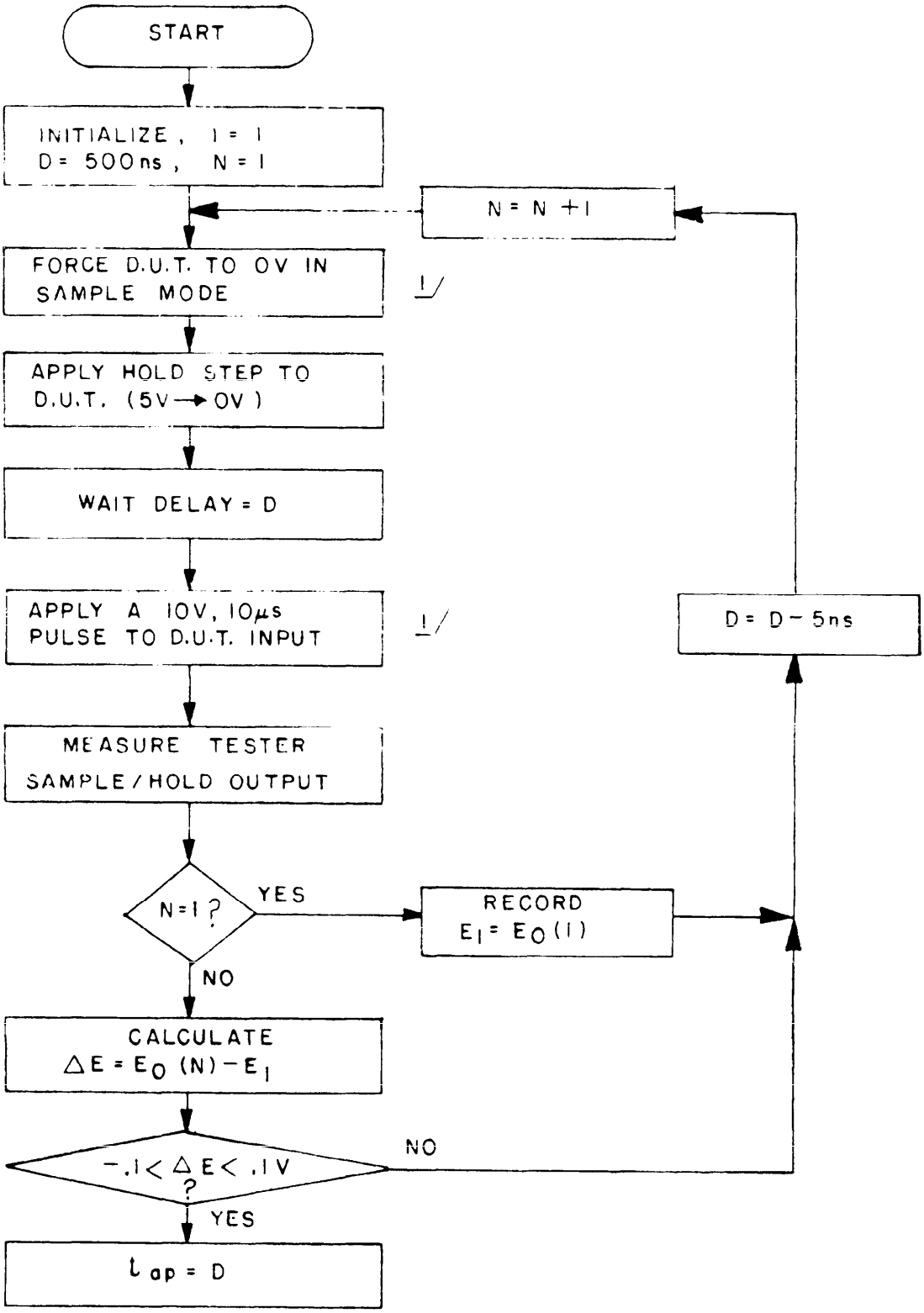
1/ Aperture time is equal to the time delay from the $V_{LOGIC\ 1}$ (DUT) 'hold' command step to the 10 V input transition which yields a 0.1 volt output error from its 2 μs delayed value $E_{O(1)}$.

2/ A flow chart for the automatic determination of aperture time is shown in Figure 17. All four combinations of 10 V input and 5 V logic signals shall be checked.

3/ The adapter S/H U2 holds the 'data' voltage for the measurement system.

FIGURE 16. Aperture time test circuit.

Source: https://assist.dla.mil -- Downloaded: 202610-21T13:28Z
Check the source to verify that this is the current version before use.



1/ Repeat above procedure for three other cases as follows:

a) Start at 0 V and apply a - 10 V pulse.
b) Start at 10 V and apply a - 10 V pulse.
c) Start at - 10 V and apply a + 10 V pulse.

FIGURE 17. Aperture time flow chart.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:28Z
Check the source to verify that this is the current version before use.

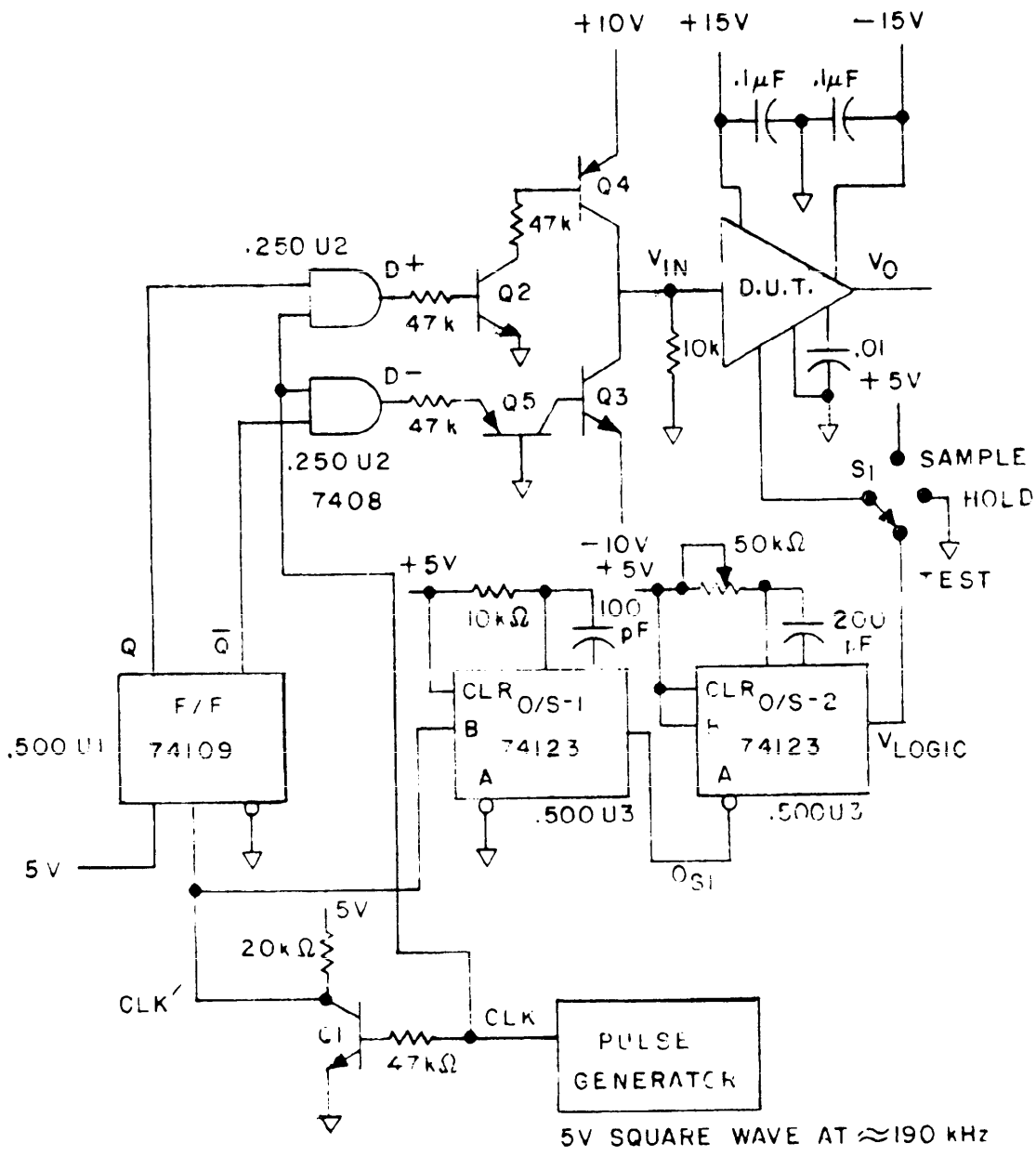
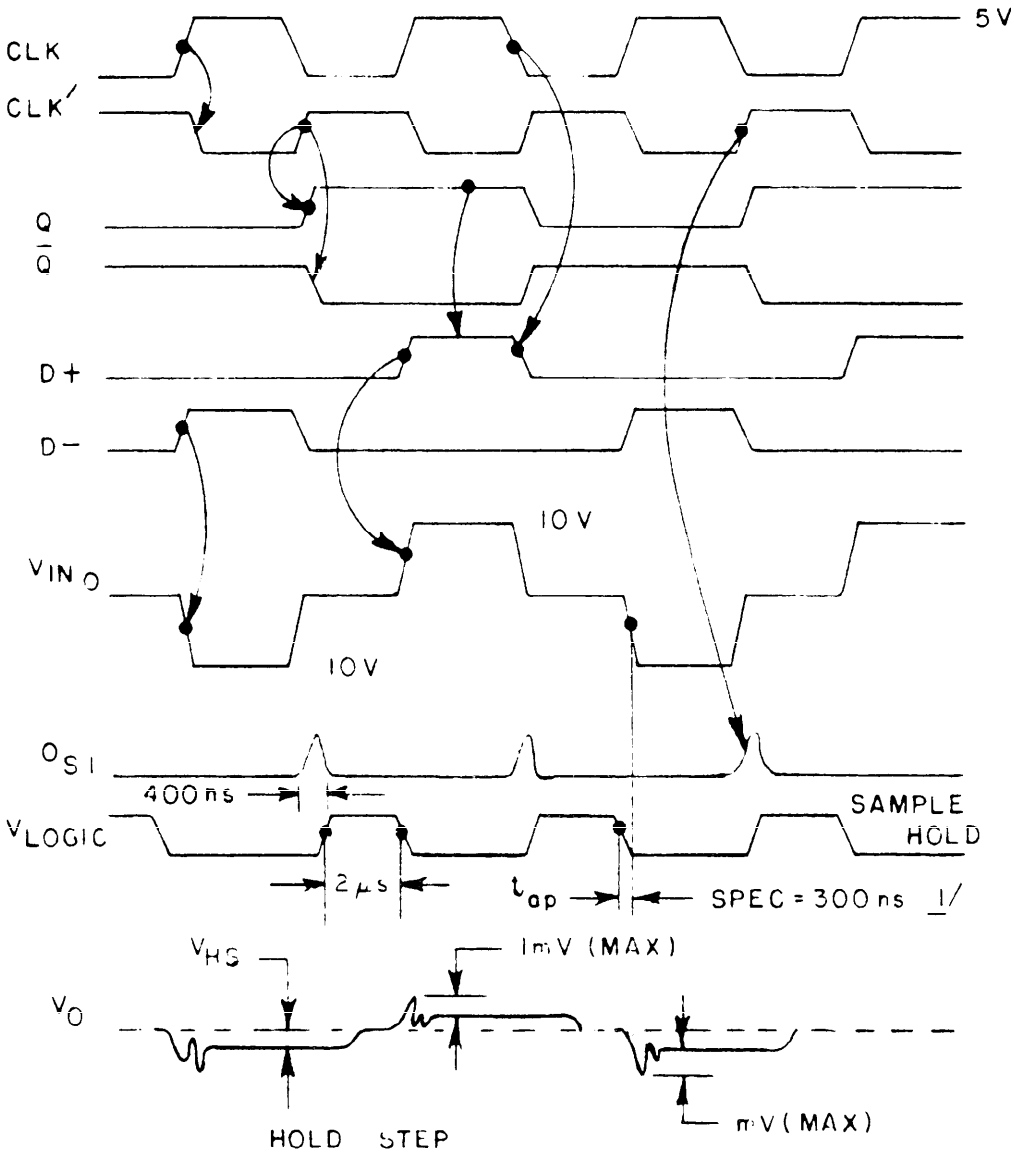


FIGURE 18. Alternate aperture time test circuit.

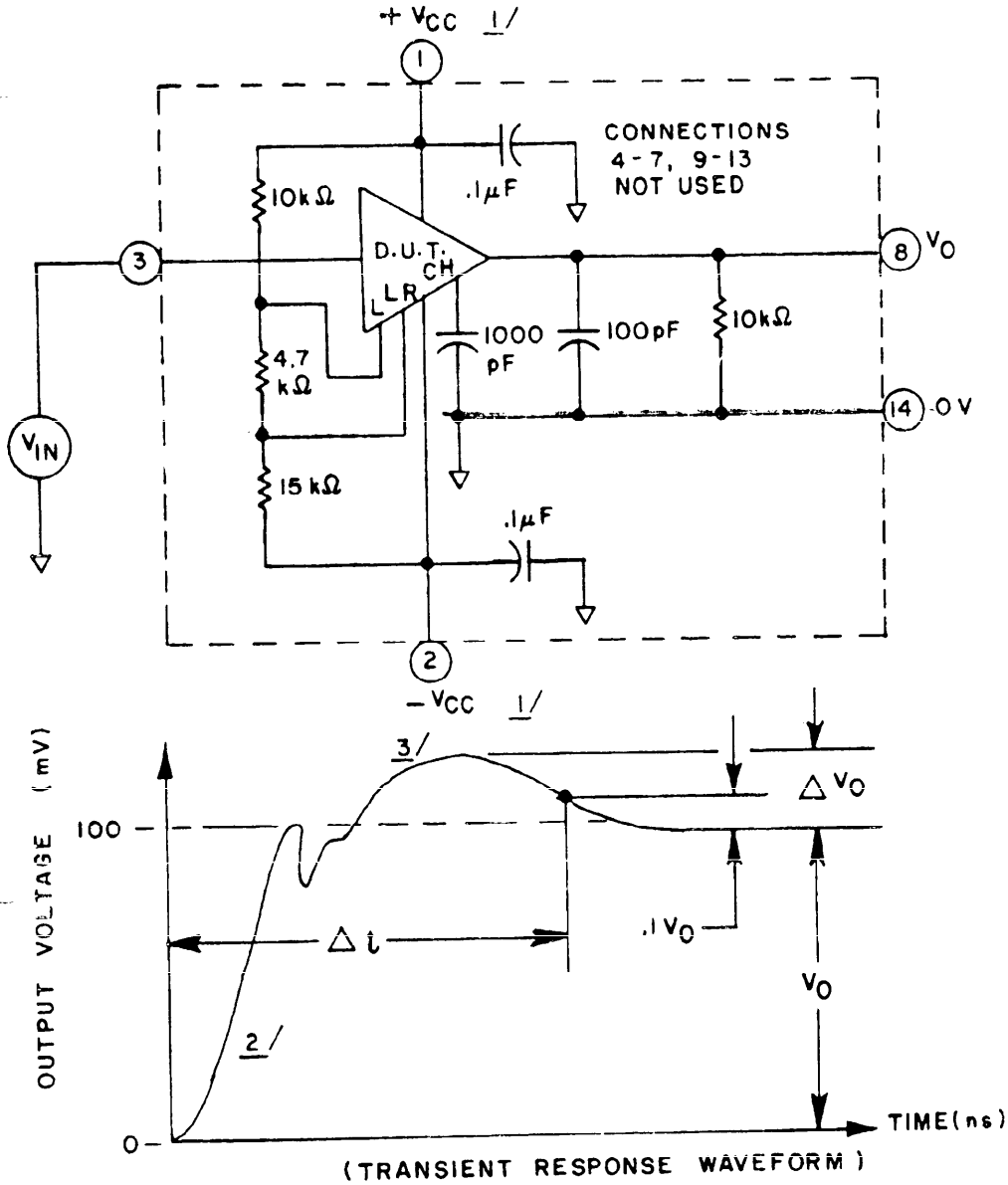
Source: https://assist.dla.mil -- Downloaded: 202610-21T13:28Z
Check the source to verify that this is the current version before use.



1/ The clock frequency and logic pulse width are adjusted so that there is a 300 nanosecond delay from the logic Hold transition to a ±10 V input transition. For these conditions the effect on the output shall be less than 1 mV. Laboratory instruments may be used to apply similar input conditions.

FIGURE 19. Alternate aperture time test waveforms.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:28Z
Check the source to verify that this is the current version before use.

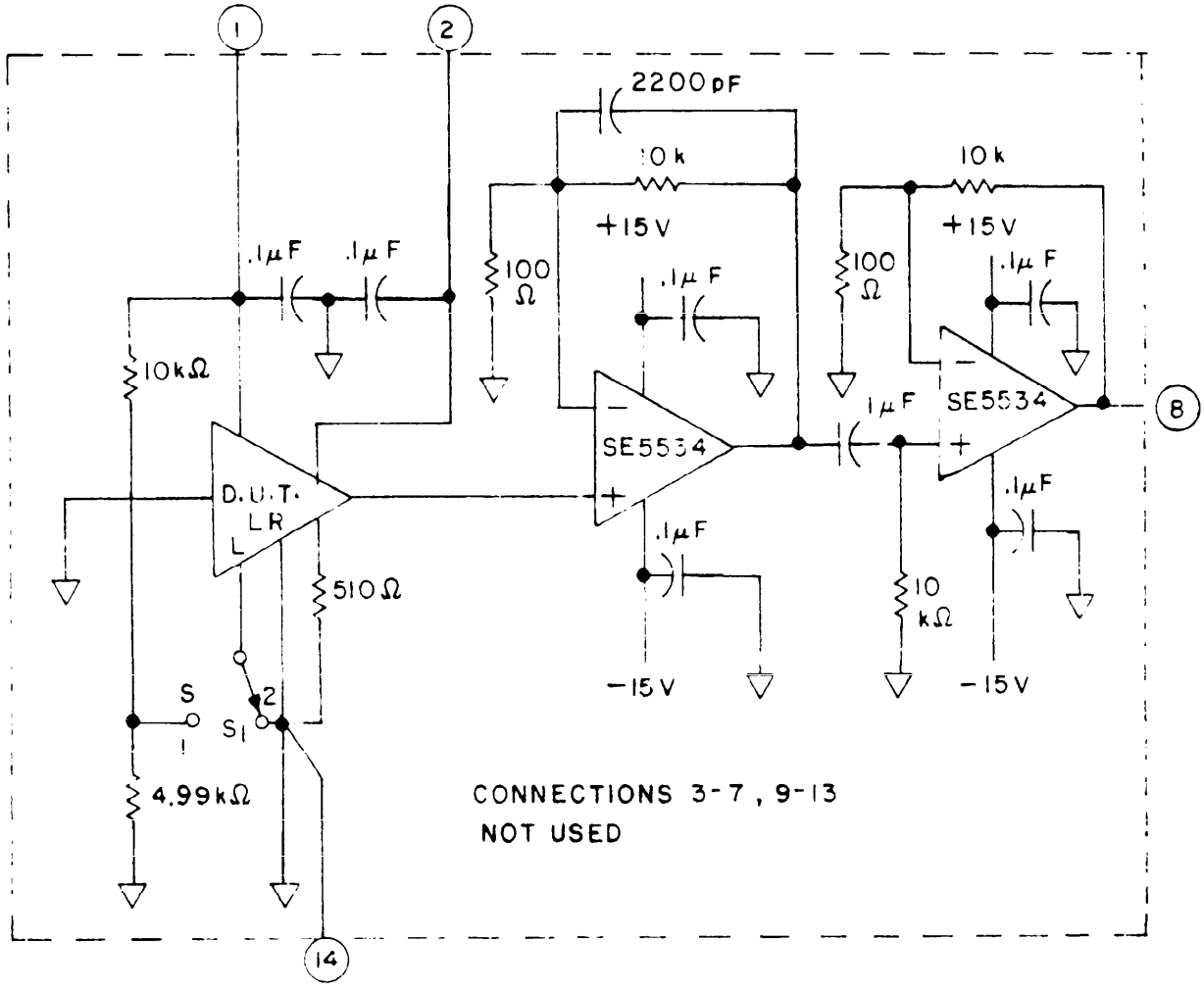


Parameter Symbol	Input pulse signal @ $t_r \leq 50$ ns	Output pulse signal (See notes)	Equation
$TR_{(ts)}$	100 mV	Above waveform	$TR_{(ts)} = \Delta t$
$TR_{(os)}$	100 mV	Above waveform	$TR_{(os)} = \Delta V_o / V_o$

NOTES:
1/ + V_{CC} and - V_{CC} shall be at the common mode limits. (i.e. first + 3.5 V, - 26.5 V and then + 26.5 V, - 3.5 V.)
2/ Any high frequency ringing shall be over within 1 μs.
3/ After its peak the major loop response shall be without further oscillations.

FIGURE 20. Transient response test circuit and waveform.

Source: https://assist.dla.mil -- Downloaded: 202610-21T13:28Z
Check the source to verify that this is the current version before use.



Symbol	S1	Measure Value	Measure Units	Input Noise Equation	Units
$e_n(S)$	1	Eo	mVrms	$e_n(S) = 0.1$ Eo	uVrms
$e_n(H)$	2	Eo	mVrms	$e_n(H) = 0.1$ Eo	uVrms

NOTES:
1/ The circuit components are designed to provide an effective "brickwall" bandwidth from 10 Hz to 10 kHz.
2/ Measurement should be made with a true rms voltmeter with at least 20 kHz bandwidth.
3/ The filter pole frequencies are related to the effective noise pass band frequencies as follows: $f_{LE} = \frac{2}{\pi} f_{LP}$; $f_{HE} = \frac{\pi}{2} f_{HP}$

FIGURE 21. Noise test circuit.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:28Z
Check the source to verify that this is the current version before use.

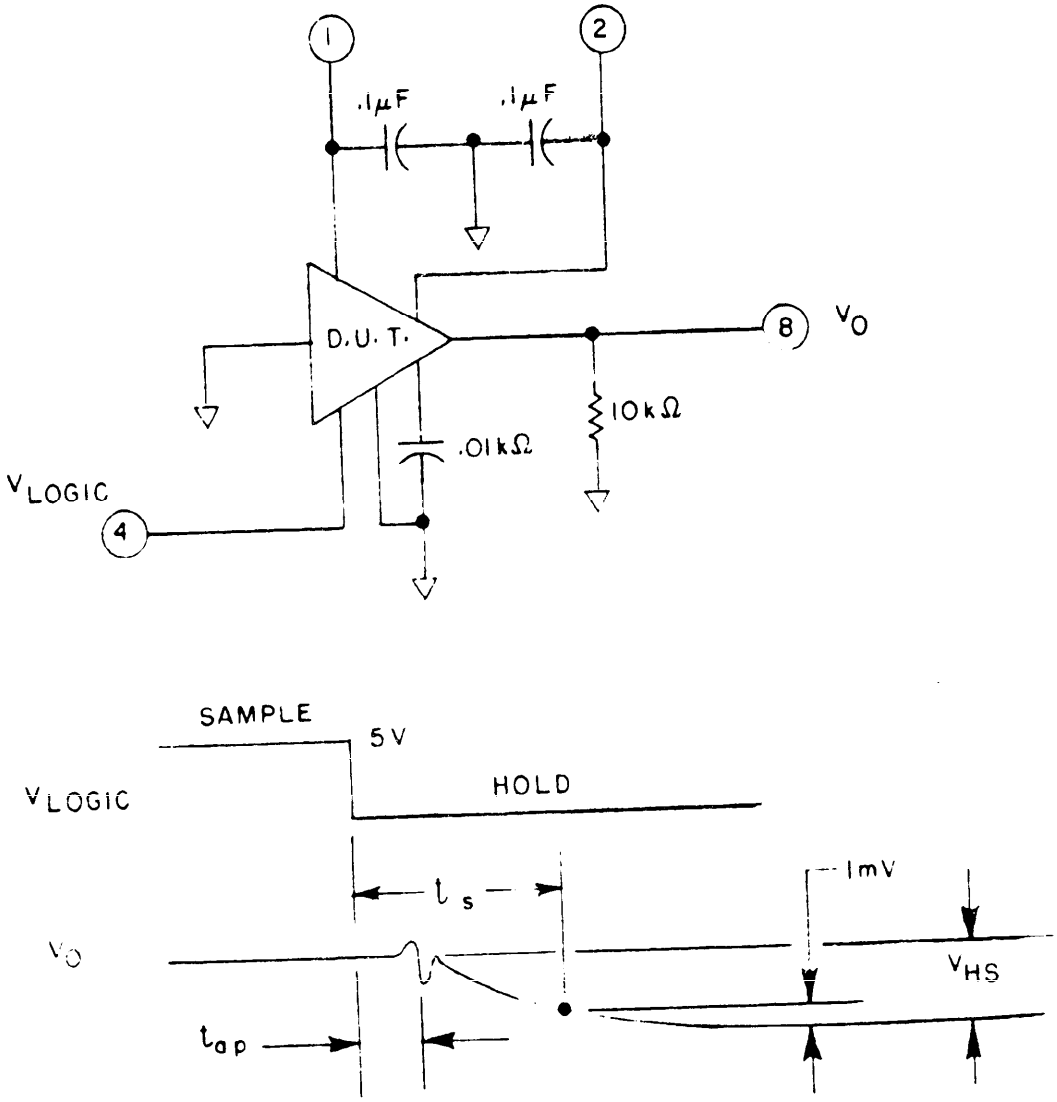
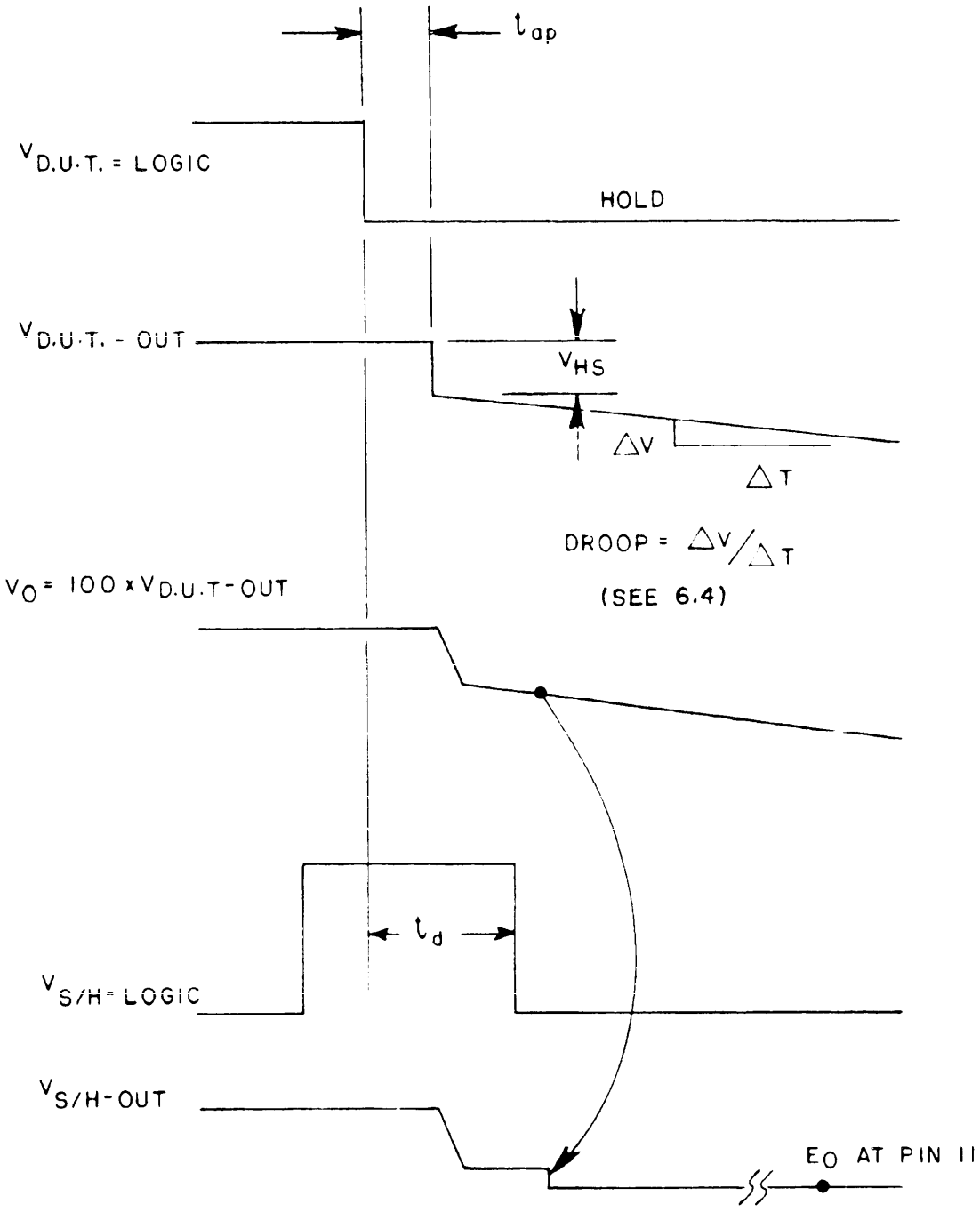


FIGURE 22. Hold mode settling time test circuit.

Source: https://assist.dla.mil -- Downloaded: 202610-21T13:28Z
Check the source to verify that this is the current version before use.



NOTES:
1/ At 125°C hold mode leakage current causes excessive droop rate of the hold capacitor voltage.
(i.e., max. droop rate = $\Delta V/\Delta t$ = I/C = 100 nA/.01 µF = 10 mV/ms)
2/ In order to minimize the droop error, the test fixture S/H at 25°C is used to acquire and hold the voltage until the measurement system can do its function.
3/ See figure 7 test circuit.

FIGURE 23. 125°C DUT Hold Mode Measurements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:28Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for all device types.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Notes	Adapter pin numbers 1	2	3	4	5	6	7	Energized relays	Measured pin No.	Value	Unit	Equation 1/	Limits Min	Max	Unit
Calibration			1	2/	Apply 0 V to pin 9 with DUT removed							None	8	E_1	V	$V_{CAL} = 10\ E_1$	-2	2	mV
1 T_A = 25°C	V_{IO}	4001	2	3/	3.5 V	-26.5 V	Open	-9 V	-11.5 V	Open	Open			E_2		$V_{IO} = 10\ (E_2-E_1)$	-3	3	
			3		26.5 V	-3.5 V		14 V	11.5 V					E_3		$V_{IO} = 10\ (E_3-E_1)$			
			4		15 V	-15 V		2.5 V	0 V					E_4		$V_{IO} = 10\ (E_4-E_1)$			
			5		7 V	-3 V		4.5 V	2 V					E_5		$V_{IO} = 10\ (E_5-E_1)$			
			6		3 V	-7 V		0.5 V	-2 V					E_6		$V_{IO} = 10\ (E_6-E_1)$			
	I_{IB}		7		3.5 V	-26.5 V		-9 V	-11.5 V			K1		E_7		$I_{IB} = 100\ (E_2-E_7)$	-1	25	nA
			8		26.5 V	-3.5 V		14 V	11.5 V					E_8		$I_{IB} = 100\ (E_3-E_8)$			
			9		15 V	-15 V		2.5 V	0 V					E_9		$I_{IB} = 100\ (E_4-E_9)$			
			10		7 V	-3 V		4.5 V	2 V					E_{10}		$I_{IB} = 100\ (E_5-E_{10})$			
			11		3 V	-7 V		0.5 V	-2 V					E_{11}		$I_{IB} = 100\ (E_6-E_{11})$			
	Z_I		12		Calculate value using data from tests 2, 3, 7, and 8											$Z_I = 0.23/\lvert E_2 + E_8 - E_3 - E_7 \rvert$	2	---	GΩ
	A_e		13		3.5 V	-26.5 V	Open	-9 V	-11.5 V	Open	Open	K4 K4	8	E_{12} E_{13}	V	$A_e = (E_{12}-E_{13})/23$	-.005	.005	V/V
			14	-02 only	5 V 25 V	-25 V -5 V		-7.5 V 12.5 V	-10 V 10 V			K7 K7		E_{14} E_{15}		$A_e = (E_{14}-E_{15})/20$			
			15		3 V 7 V	-7 V -3 V		0.5 V 4.5 V	-2 V 2 V			K4 K4		E_{16} E_{17}		$A_e = (E_{16}-E_{17})/4$	-.02	.02	
	$V_{IO(ADJ+)}$		16		15 V	-15 V		2.5 V	0 V			K5 K8		E_{18} E_{19}		$V_{IO(ADJ+)} = 10\ (E_4-E_{18})$ $V_{IO(ADJ-)} = 10\ (E_4-E_{19})$	6 ---	--- -6	mV mV
	$V_{IO(ADJ-)}$		17		15 V	-15 V		2.5 V	0 V										
	+PSRR	4003	18		18 V 12 V	-18 V -18 V		2.5 V -0.5 V	0 V -3 V			None None		E_{20} E_{21}		$+PSRR = 20\ \log\ \lvert 600/(E_{20}-E_{21}) \rvert$	80	---	dB
	-PSRR	4003	19		18 V	-12 V		5.5 V	3 V			None		E_{22}		$-PSRR = 20\ \log\ \lvert 600/(E_{20}-E_{22}) \rvert$	80		
	FRR		20 21 22 23	4/,5/	15 V	-15 V	See fig. 8 timing		0 V			K1, K2		E_{23} E_{24} E_{25} E_{26} E_{27}		$FRR = 20\ \log\ \lvert 1150/(E_{24}-E_{23}) \rvert$ $FRR = 20\ \log\ \lvert 1150/(E_{25}-E_{24}) \rvert$ $FRR = 20\ \log\ \lvert 1150/(E_{26}-E_{25}) \rvert$ $FRR = 20\ \log\ \lvert 1150/(E_{27}-E_{26}) \rvert$	86		
	Z_O		24	6/			See fig. 10 timing			-10 V 10 V	0 V 0 V	K3		E_{28} E_{29}		$Z_O = 5\ (E_{29}-E_{28})$	---	2	Ω
	V_{HS}		25	5/,7/	3.5 V	-26.5 V	See fig. 11 timing		-11.5 V	Open	Open	K6		E_{30} E_{31}		$V_{HS} = 10\ (E_{31}-E_{30})$	-2		mV
			26		26.5 V	-3.5 V			11.5 V					E_{32} E_{33}		$V_{HS} = 10\ (E_{33}-E_{32})$			
	I_{CC}		27		15 V	-15 V	Open	2.5 V	0 V			None	2	I_1	mA	$I_{CC} = I_1$ Type 01 Type 02	1 1	5.5 6.0	mA mA
	R_{SC}		28		15 V	-15 V	0 V	2.5 V	0 V		0 V	K1, K2, K3	7	I_2	mA	$R_{SC} = 400/(I_3-I_2)$	100	400	Ω
					15 V	-15 V	0.4 V	2.5 V	0 V		0 V	K1, K2, K3	7	I_3	mA				
	I_{IH}		29	8/	8.5 V	-21.5 V	Open	5.5 V	0 V		Open	None	4	I_4	µA	$I_{IH} = I_4$	0	10	µA
	I_{IH}		30		8.5 V	-21.5 V		0 V	5.5 V				5	I_5		$I_{IH} = I_5$	0	10	
	I_{IL}		31		21.5 V	-8.5 V		0 V	0 V				4	I_6		$I_{IL} = I_6$	-1.0	1.0	
	I_{IL}		32		21.5 V	-8.5 V		0 V	0 V				5	I_7		$I_{IL} = I_7$	-1.0	1.0	

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-02-21T13:28Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for all device types - Continued.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Notes	Adapter pin numbers 1	2	3	4	5	6	7	Energized relays	Measured pin No.	Value	Unit	Equation 1/	Limits Min	Max	Unit
1 T_A = 25°C	$I_{OS(+)}$	3011	33	9/	15 V	-15 V	10 V	2.5 V	0 V	Open	Open	K1, K2	9	I_8	mA	$I_{OS(+)} = I_8$ Type 01	-25	---	mA
																Type 02	-30	---	mA
	$I_{OS(-)}$	3011	34	9/	15 V	-15 V	-10 V	2.5 V	0 V			K1, K2	9	I_9	mA	$I_{OS(-)} = I_9$ Type 01		25	V
																Type 02	---	30	V
	$I_{HL(+)}$		35	10/	3.5 V	-26.5 V	See fig. 12 timing (at = 100 ms)		-11.5 V			None	8	E_{34} E_{35}	mV	$I_{HL(+)} = (E_{35}-E_{34})$	-100	100	pA
	$I_{HL(-)}$		36	10/	26.5 V	-3.5 V			11.5 V					E_{36} E_{37}		$I_{HL(-)} = (E_{37}-E_{36})$			
	$I_{CH(+)}$		37	11/	15 V	-15 V	11.5 V	2.5 V	0 V		9.5 V	K1, K2, K3	7	I_{10}	mA	$I_{CH(+)} = I_{10}$	---	-3	mA
	$I_{CH(-)}$		38	11/			-11.5 V	2.5 V			-9.5 V			I_{11}	mA	$I_{CH(-)} = I_{11}$	3	---	mA
	$V_{TH(H)}$		39	12/			-2 V	2.0 V			0 V			I_{12}	mA	$V_{TH(H)} \leq 2$ V if $I_{12} \geq 1$ mA	1	---	mA
	$V_{TH(L)}$		40	12/			-2 V	0.8 V			0 V			I_{13}	µA	$V_{TH(L)} \leq 0.5$ V if $-10 < I_{13} < 10$ µA	-10	10	µA
2 T_A = 125°C	V_{IO}	4001	41	2/,3/	3.5 V	-26.5 V	Open	-9 V	-11.5 V		Open	None	8	E_{38}	V	$V_{IO} = 10 (E_{38}-E_1)$	-5	5	mV
			42		26.5 V	-3.5 V		14 V	11.5 V					E_{39}		$V_{IO} = 10 (E_{39}-E_1)$			
			43		15 V	-15 V		2.5 V	0 V					E_{40}		$V_{IO} = 10 (E_{40}-E_1)$			
			44		7 V	-3 V		4.5 V	2 V					E_{41}		$V_{IO} = 10 (E_{41}-E_1)$			
			45		3 V	-7 V		0.5 V	-2 V					E_{42}		$V_{IO} = 10 (E_{42}-E_1)$			
	I_{IB}		46	3/	3.5 V	-26.5 V		9 V	-11.5 V			K1		E_{43}		$I_{IB} = 100 (E_{38}-E_{43})$	-25	75	nA
			47		26.5 V	-3.5 V		14 V	11.5 V					E_{44}		$I_{IB} = 100 (E_{39}-E_{44})$			
			48		15 V	-15 V		2.5 V	0 V					E_{45}		$I_{IB} = 100 (E_{40}-E_{45})$			
			49		7 V	-3 V		4.5 V	2 V					E_{46}		$I_{IB} = 100 (E_{41}-E_{46})$			
			50		3 V	-7 V		0.5 V	-2 V					E_{47}		$I_{IB} = 100 (E_{42}-E_{47})$			
	Z_I		51		Calculate value using data from tests 41,42,46, and 47											$Z_I = 0.23/\|E_{38}+E_{44}-E_{39}-E_{43}\|$	1	---	GΩ
	A_e		52		3.5 V	-26.5 V	Open	-9 V	-11.5 V	Open	Open	K4	8	E_{48}	V	$A_e = (E_{48}-E_{49})/23$	.02	.02	V/V
					26.5 V	-3.5 V		14 V	11.5 V			K4		E_{49}					
			53	-02 only	5 V	-25 V		-7.5 V	-10 V			K7		E_{50}		$A_e = (E_{50}-E_{51})/20$			
					25 V	-5 V		12.5 V	10 V			K7		E_{51}					
			54		3 V	-7 V		0.5 V	-2 V			K4		E_{52}		$A_e = (E_{52}-E_{53})/4$	.04	.04	
					7 V	-3 V		4.5 V	2 V			K4		E_{53}					
	$V_{IO(ADJ+)}$		55		15 V	-15 V		2.5 V	0 V			K5		E_{54}		$V_{IO(ADJ+)} = 10 (E_{40}-E_{54})$	6	---	mV
	$V_{IO(ADJ-)}$		56		15 V	-15 V		2.5 V	0 V			K8		E_{55}		$V_{IO(ADJ-)} = 10 (E_{40}-E_{55})$	---	-6	mV
	+PSRR	4003	57		18 V	-18 V		2.5 V	0 V			None		E_{56}		+PSRR = 20 log |600/($E_{56}-E_{57}$)|	80	---	dB
					12 V	-18 V		0.5 V	-3 V			None		E_{57}					
	-PSRR	4003	58		18 V	-12 V		5.5 V	3 V			None		E_{58}		-PSRR = 20 log |600/($E_{56}-E_{58}$)|			
	FRR		59	4/,5/ 13/	15 V	-15 V	See fig. 8 timing waveforms and fig. 23		0 V			K1, K2	11	E_{59}		FRR = 20 log |1150/($E_{60}-E_{59}$)|			
			60											E_{60}		FRR = 20 log |1150/($E_{61}-E_{60}$)|			
			61											E_{61}		FRR = 20 log |1150/($E_{62}-E_{61}$)|			
			62											E_{62} E_{63}		FRR = 20 log |1150/($E_{63}-E_{62}$)|			

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-02-21T13:28Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for all device types - Continued.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Notes	Adapter pin numbers 1	2	3	4	5	6	7	Energized relays	Measured pin No.	Value	Unit	Equation 1/	Limits Min	Max	Unit
2 T_A = 125°C	Z_O		63	6/,13/	15 V 15 V	-15 V -15 V	See fig. 10 and fig. 23		0 V 0 V	-10 V 10 V	10 V 10 V	K3 K3	11	E_{64} E_{65}	V	$Z_O = 5\ (E_{65}-E_{64})$	---	2	Ω
	V_{HS}		64	5/,7/, 13/	3.5 V	-26.5 V	See fig. 11 and fig. 23		-11.5 V	Open	Open	K6		E_{66} E_{67}		$V_{HS} = 10\ (E_{67}-E_{66})$	-5	5	mV
			65		26.5 V	-3.5 V			11.5 V					E_{68} E_{69}		$V_{HS} = 10\ (E_{69}-E_{68})$			
	I_{CC}		66		15 V	-15 V	Open	2.5 V	0 V			None	2	I_{14}	mA	$I_{CC} = I_{14}$ Type 01 Type 02	1 1	15.5 16.0	mA mA
	R_{SC}		67		15 V 15 V	-15 V -15 V	0 V 0.4 V	2.5 V 2.5 V	0 V 0 V		0 V 0 V	K1, K2, K3	7	I_{15} I_{16}	mA mA	$R_{SC} = 400/(I_{16}-I_{15})$	100	400	Ω
	I_{IH} I_{IH}		68 69		8.5 V	-21.5 V	Open	5.5 V 0 V	0 V 5.5 V		Open	None	4 5	I_{17} I_{18}	μA	$I_{IH} = I_{17}$ $I_{IH} = I_{18}$	0 0	10 10	μA
	I_{IL} I_{IL}		70 71		21.5 V	-8.5 V		0 V 0 V	0 V				4 5	I_{19} I_{20}		$I_{IL} = I_{19}$ $I_{IL} = I_{20}$	-1 -1	1 1	
	$I_{OS(+)}$	3011	72	9/	15 V	-15 V	10 V	2.5 V				K1, K2	9	I_{21}	mA	$I_{OS(+)} = I_8$ Type 01 Type 02	-25 -30	--- ---	mA
	$I_{OS(-)}$	3011	73	9/	15 V	-15 V	-10 V	2.5 V				K1, K2	9	I_{22}	mA	$I_{OS(-)} = I_9$ Type 01 Type 02	--- ---	25 30	
	$I_{HL(+)}$		74	10/,13/	3.5 V	-26.5 V	See fig. 12 and fig. 23 (at = 10 ms)		-11.5 V			None	11	E_{70} E_{71}	mV	$I_{HL(+)} = 0.1\ (E_{71}-E_{70})$	-50	50	nA
	$I_{HL(-)}$		75		26.5 V	-3.5 V			11.5 V					E_{72} E_{73}		$I_{HL(-)} = 0.1\ (E_{73}-E_{72})$			
	$I_{CH(+)}$ *		76	11/	15 V	-15 V	11.5 V	2.5 V	0 V		8.5 V	K1,K2,K3	7	I_{23}	mA	$I_{CH(+)} = I_{23}$	---	-2	mA
	$I_{CH(-)}$		77	11/			-11.5 V	2.5 V			-9.5 V			I_{24}	mA	$I_{CH(-)} = I_{24}$	2	---	
	$V_{TH(H)}$		78	12/			-2 V	2.0 V			0 V			I_{25}	mA	$V_{TH(H)} \leq 2$ V if $I_{25} \geq 1$ mA	1	---	
	$V_{TH(L)}$		79	12/			-2 V	0.8 V			0 V			I_{26}	μA	$V_{TH(L)} \geq 0.8$ V if $-10 \leq I_{26} < 10$ μA	-10	10	μA
3 T_A = -55°C	V_{IO}	4001	80 81 82 83 84	3/	3.5 V 26.5 V 15 V 7 V 3 V	-26.5 V -3.5 V -15 V -3 V -7 V	Open	-9 V 14 V 2.5 V 4.5 V 0.5 V	-11.5 V 11.5 V 0 V 2 V -2 V		Open	None	8	E_{74} E_{75} E_{76} E_{77} E_{78}	V	$V_{IO} = 10\ (E_{74}-E_1)$ $V_{IO} = 10\ (E_{75}-E_1)$ $V_{IO} = 10\ (E_{76}-E_1)$ $V_{IO} = 10\ (E_{77}-E_1)$ $V_{IO} = 10\ (E_{78}-E_1)$	-5	5	mV
	I_{IB}		85 86 87 88 89		3.5 V 26.6 V 15 V 7 V 3 V	-26.5 V -3.5 V -15 V -3 V -7 V		-9 V 14 V 2.5 V 4.5 V 0.5 V	-11.5 V 11.5 V 0 V 2 V -2 V			K1		E_{79} E_{80} E_{81} E_{82} E_{83}		$I_{IB} = 100\ (E_{74}-E_{79})$ $I_{IB} = 100\ (E_{75}-E_{80})$ $I_{IB} = 100\ (E_{76}-E_{81})$ $I_{IB} = 100\ (E_{77}-E_{82})$ $I_{IB} = 100\ (E_{78}-E_{83})$	-25	75	nA
	Z_I		90		Calculate value using data from tests 80,81,85, and 86											$Z_I = 0.23/\ \lvert E_{74}+E_{80}-E_{75}-E_{79}\rvert$	1	---	GΩ
	A_e		91		3.5 V 26.5 V	-26.5 V -3.5 V	Open	-9 V 14 V	-11.5 V 11.5 V	Open	Open	K4 K4	8	E_{84} E_{85}	V	$A_e = (E_{84}-E_{85})/23$	-.02	.02	%
			92	-02 only	5 V 25 V	-25 V -5 V		-7.5 V 12.5 V	-10 V 10 V			K7 K7		E_{86} E_{87}		$A_e = (E_{86}-E_{87})/20$			
			93		3 V 7 V	-7 V -3 V		0.5 V 4.5 V	-2 V 2 V			K4 K4		E_{88} E_{89}		$A_e = (E_{88}-E_{89})/4$	-.04	.04	

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-02-21T13:28Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for all device types - Continued.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Notes	Adapter pin numbers 1	2	3	4	5	6	7	Energized relays	Measured pin No.	Value	Unit	Equation 1/	Limits Min	Max	Unit
3 T_A = -55°C	$V_{IO(ADJ+)}$		94		15 V	-15 V	Open	2.5 V	0 V	Open	Open	K5	8	E_{90}	V	$V_{IO(ADJ+)} = 10\ (E_{76}-E_{90})$	6	---	mV
	$V_{IO(ADJ-)}$		95		15 V	-15 V		2.5 V	0 V			K8		E_{91}		$V_{IO(ADJ-)} = 10\ (E_{76}-E_{91})$	---	-6	mV
	+PSRR	4003	96		18 V 12 V	-18 V -18 V		2.5 V -0.5 V	0 V -3 V			None None		E_{92} E_{93}		$+PSRR = 20 \log \lvert 600/(E_{92}-E_{93})\rvert$	80	---	dB
	-PSRR	4003	97		18 V	-12 V		5.5 V	3 V			None		E_{94}		$-PSRR = 20 \log \lvert 600/(E_{92}-E_{94})\rvert$			
	FRR		98 99 100 101	4/,5/	15 V	-15 V	See fig. 8 timing		0 V			K1, K2		E_{95} E_{96} E_{97} E_{98} E_{99}		$FRR = 20 \log \lvert 1150/(E_{96}-E_{95})\rvert$ $FRR = 20 \log \lvert 1150/(E_{97}-E_{96})\rvert$ $FRR = 20 \log \lvert 1150/(E_{98}-E_{97})\rvert$ $FRR = 20 \log \lvert 1150/(E_{99}-E_{98})\rvert$			
	Z_O		102	6/			See fig. 10			-10 V 10 V	0 V 0 V	K3 K3		E_{100} E_{101}		$Z_O = 5\ (E_{101}-E_{100})$	---	2	Ω
	V_{HS}		103	5/,7/	3.5 V 3.5 V	-26.5 V -26.5 V	See fig. 11		-11.5 V -11.5 V	Open	Open	K6		E_{102} E_{103}		$V_{HS} = 10\ (E_{103}-E_{102})$	-5	5	mV
			104		26.5 V 26.5 V	-3.5 V -3.5 V			11.5 V 11.5 V					E_{104} E_{105}		$V_{HS} = 10\ (E_{105}-E_{104})$			mV
	I_{CC}		105		15 V	-15 V	Open	2.5 V	0 V			None	2	I_{27}	mA	$I_{CC} = I_{27}$ Type 01 Type 02	1 1	6.5 7.0	mA mA
	R_{SC}		106		15 V 15 V	-15 V -15 V	0 V 0.4 V	2.5 V 2.5 V	0 V 0 V		0 V 0 V	K1, K2, K3 K1, K2, K3	7 7	I_{28} I_{29}	mA mA	$R_{SC} = 400/(I_{29}-I_{28})$	100	400	Ω
	I_{IH}		107 108	1/	8.5 V 8.5 V	-21.5 V -21.5 V	Open	5.5 V 0 V	0 V 5.5 V		Open	None	4 5	I_{30} I_{31}	µA	$I_{IH} = I_{30}$ $I_{IH} = I_{31}$	0 0	10 10	µA
	I_{IL}		109 110		21.5 V 21.5 V	-8.5 V -8.5 V		0 V 0 V	5 V				4 5	I_{32} I_{33}		$I_{IL} = I_{32}$ $I_{IL} = I_{33}$	-1 -1	1 1	
	$I_{OS(+)}$	3011	111	9/	15 V	-15 V	0 V	2.5 V				K1, K2	9	I_{34}	mA	$I_{OS(+)} = I_{34}$ Type 01 Type 02	-25 -30	---	mA
	$I_{OS(-)}$	3011	112	9/			-10 V					K1, K2	9	I_{35}		$I_{OS(-)} = I_{35}$ Type 01 Type 02	--- ---	25 30	
	$I_{CH(+)}$		113	11/			11.5 V				9.5 V	K1, K2, K3	7	I_{36}		$I_{CH(+)} = I_{36}$	---	-2	
	$I_{CH(-)}$		114	11/			-11.5 V				-9.5 V			I_{37}		$I_{CH(-)} = I_{37}$	2	---	
	$V_{TH(H)}$		115	12/			-2 V	2.0 V			0 V			I_{38}		$V_{TH(H)} \leq 2$ V if $I_{38} \geq 1$ mA	1	---	
	$V_{TH(L)}$		116	12/			-2 V	0.8 V			0 V			I_{39}		$V_{TH(L)} \geq 0.8$ V if $-10 \leq I_{39} \leq 10$ µA	-10	10	µA
7 T_A = 25°C	t_{aq}		117 118 119 120	14/,15/			See fig. 13, 14, and 15				Open	K1, K2, K6	11	t_1 t_2 t_3 t_4	µs	$t_{aq} = t_1$ $t_{aq} = t_2$ $t_{aq} = t_3$ $t_{aq} = t_4$	---	25	µs
	t_{ap}		121 122 123 124	15/,16/			See fig. 16 thru 19							t_5 t_6 t_7 t_8	ns	$t_{ap} = t_5$ $t_{ap} = t_6$ $t_{ap} = t_7$ $t_{ap} = t_8$		300	ns
	t_s		125	17/			See fig. 22					None	8	t_9	µs	$t_s = t_9$		1.5	µs
	FRR_{ac}		126	18/			See fig. 9 V_{IN} = 20 V_{pp}					None	8	E_{106}	mV_{pp}	$FRR_{ac} = 20 \log \left\lvert \frac{20000}{E_{106}} \right\rvert$	86	---	dB

See footnotes at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-02-21T13:28Z
Check the source to verify that this is the current version before use.

TABLE III. Group A inspection for all device types - Continued.

Subgroup	Symbol	MIL-STD-883 method	Test no.	Notes	Adapter pin numbers 1	2	3	4	5	6	7	Energized relays	Measured pin No.	Value	Unit	Equation 1/	Limits Min	Max	Unit
7 T_A = 25°C	$TR_{(ts)}$		127	19/	3.5 V	-26.5 V	See fig. 20 V_{IN} = 100 mV pulse			Open	Open	None	8	t_{10}	µs	$TR_{(ts)} = \Delta t = t_{10}$	---	2.5	µs
			128		26.5 V	-3.5 V								t_{11}	µs	$TR_{(ts)} = \Delta t = t_{11}$		2.5	µs
	$TR_{(os)}$		129		3.5 V	-26.5 V								V_{01}, ΔV_{01}	mV	$TR_{(os)} = 100\ \Delta V_{01}/V_{01}$		40	%
			130		26.5 V	-3.5 V								V_{02}, ΔV_{02}	mV	$TR_{(os)} = 100\ \Delta V_{02}/V_{02}$		40	%
	$e_{n(S)}$		131	20/	15 V	-15 V	See fig. 21							E_{107}	mVrms	$e_{n(S)} = 0.1\ E_{107}$		10	mVrms
	$e_{n(H)}$		132	20/	15 V	-15 V	See fig. 21							E_{108}	mVrms	$e_{n(H)} = 0.1\ E_{108}$		10	mVrms
8 T_A = 125°C	$\frac{\Delta V_{IO}}{\Delta T}$		133	$\frac{\Delta V_{IO}}{\Delta T} = [V_{IO}$ (test 43) - V_{IO} (test 4)]/100°C													-20	20	µV/°C
8 T_A = -55°C	$\frac{\Delta V_{IO}}{\Delta T}$		134	$\frac{\Delta V_{IO}}{\Delta T} = [V_{IO}$ (test 82) - V_{IO} (test 4)]/80°C													-20	20	µV/°C

1/ The equations take into account the test amplifier gain of 100 and other circuit constants so that the calculated value is in table I units.

2/ In order to remove test amplifier offset from the data values measure the offset of U1 on pin 8 with pin 9 grounded and the DUT removed. Software subtraction techniques shall be used to correct the data.

3/ Common mode input range conditions are exercised by grounding the signal input and swinging the power supplies to their nominal levels minus the common mode voltage. For example for V_{CM} = +11.5 V, $+V_{CC}$ = 15 V - 11.5 = 3.5 V and $-V_{CC}$ = -15 V - 11.5 V = -26.5 V.

4/ With a 0 V signal input the DUT logic input is switched from 2.5 V to 0 V. This resets the system in the hold mode. The test amplifier output is measured immediately after each 11.5 V change at the signal output.

5/ Logic input step changes should have a rise time of 0.5 µs or less.

6/ E28 and E29 are measured with the DUT in the hold mode and with the hold capacitor terminal grounded.

7/ For the hold step test, the first and second measurements are made with the DUT in the sample and hold modes, respectively. The hold measurement should be made within 50 µs of the DUT hold command, especially at 125°C.

8/ High and low state logic input currents shall be measured over the common mode voltage range as shown.

9/ The output shall be shorted to ground for 25 ms or less.

10/ Hold leakage current at 25°C is determined by measuring the droop referred to the test amplifier output over a 100 ms interval.

11/ The charge current measurements on pin 7 is referenced to forced voltage of 9.5 V and -9.5 V, respectively.

12/ With worst case logic threshold voltages applied, the hold cap. terminal output current is measured to determine if the device is in the correct operating mode. The logic threshold levels for $V_{TH(H)}$ and $V_{TH(L)}$ are guaranteed by measuring the hold capacitor current.

For $V_{TH(H)}$ hold capacitor current $\geq$1 mA.

For $V_{TH(L)}$ hold capacitor current $\leq$10 µA.

13/ Hold mode droop at 125°C due to high JFET leakage current will tend to mask the data for V_{HS}, Z_O, and FRR. To preclude this effect from happening, the test adapter sample/hold circuit, which is not at an elevated temperature, should be used to acquire and hold the data for the measurement system.

14/ Step the signal input from 0 V to +10 V. After a delay of $\approx$ 100 µs, generate a 100 µs sample mode pulse. The difference between the input and DUT output is monitored with a 100 V/V differential amplifier followed by tester sample/hold circuit. Reduce the DUT sample mode pulse width until there is a 100 mV (.01%) change at the tester sample/hold output from the 100 µs pulse value. The sample mode pulse width for this condition is the acquisition time. Repeat the above procedure for input signal transitions of 10 V to 0 V, 0 V to -10 V, and -10 V to 0 V. Figure 14 shows an automatic flow chart method and figure 15 shows a simplified manual method for determining acquisition time.

15/ Even with a Teflon hold capacitor servicing the DUT sample hold circuit, dielectric absorption errors can occur when dynamic signals are applied to the DUT input. To minimize these errors, the error amplifier output should be measured immediately after the test event. The tester sample and hold circuit is used to hold the analog data for the slower responding automatic measurement system.

16/ Step the logic input from 5 V to 0 V with the input at 0 V. After a 2 µs time delay, step the signal input up to 10 V. For this condition the DUT output should be $\approx$ 0 V. Gradually decrease the delay until a 100 mV (.01 %) shift occurs at the tester sample/hold output. Repeat the above procedure for input signal transitions of 10 V to 0 V, 0 V to -10 V, and -10 V to 0 V. The delay corresponding to the 100 mV shift is equal to the aperture time. See figures 16 thru 19.

17/ Settling time is determined as shown on figure 22 as the time for the S/H to settle within 1 mV of final value after the hold command is given.

18/ Dynamic feedthrough rejection is determined in the hold mode with a signal input of 20 V_{pp} at a frequency of 1 kHz.

19/ Overshoot $TR_{(os)}$ and settling time $TR_{(ts)}$ are indicative of the stability of the device.

20/ Broadband noise $e_{n(S)}$ and $e_{n(H)}$ is measured with a 100 V/V low noise bandpass amplifier as shown on figure 21.

Source: https://assist.dla.mil -- Downloaded: 2026-02-21T13:28Z
Check the source to verify that this is the current version before use.

4.5.2 Life test cooldown procedure. When devices are measured at 25°C following application of the operating life or burn-in test condition, they shall be cooled to room temperature prior to removal of the bias.

TABLE IV. Group C end-point electrical parameters.
(T_A = 25°C, $\pm V_{CC}$ = ±15 V for all device types)

Table III test no.	Test	Limits		Delta		Unit
		Min	Max	Min	Max	
4	V_{IO}	-3	3	-.5	.5	mV
9	I_{IB}	-1	25	-2.5	2.5	nA

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-38510.

6. NOTES

6.1 Notes. The notes specified in MIL-M-38510 are applicable to this specification.

6.2 Intended use. Microcircuits conforming to this specification are intended for original equipment design applications and logistic support of existing equipment.

6.3 Ordering data. The contract or purchase order should specify the following:

a. Complete part number (see 1.2).

b. Requirements for delivery of one copy of the quality conformance inspection data pertinent to the device inspection lot to be supplied with each shipment by the device manufacturer, if applicable.

c. Requirements for certificate of compliance, if applicable.

d. Requirements for notification of change of product or process to contracting activity in addition to notification to the qualifying activity, if applicable.

e. Requirements for packaging and packing.

f. Requirements for failure analysis (including required test condition of method 5003 of MIL-STD-883), corrective action, and reporting of results, if applicable.

g. Requirements for product assurance options.

h. Requirements for special carriers, lead lengths, or lead forming, if applicable. These requirements shall not affect the part number. Unless otherwise specified, these requirements will not apply to direct purchase by or direct shipment to the Government.

6.4 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-M-38510, MIL-STD-1331, and as follows:

A_E - - - - - - - - Gain error. The ratio of "sample" mode output voltage swing to input common mode voltage swing expressed in percent. To the user this can be interpreted as the percentage deviation from unity gain.

DR(+), DR(-) - - - Droop rate. The rate of change of hold capacitor voltage with time due to "hold" mode leakage current. Note that $I_{HL} = C_H \cdot \Delta V_{HC}/\Delta t = CH \cdot DR$.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:28Z
Check the source to verify that this is the current version before use.

SE - - - - - - - -	Dynamic sampling error. The error introduced into the held output due to a changing analog input when the "hold" command is given. This error is proportional to the product of input signal slew rate, hold capacitance, and the series charge resistor.
$en_{(S)}$, $en_{(H)}$ - -	Noise. The total rms noise of the device that exists within a 10 Hz to 10 kHz "brickwall" bandwidth. Both "sample" mode, $en_{(S)}$, and "hold" mode, $en_{(H)}$, specifications exist.
FRR - - - - - - - -	Feedthrough rejection ratio. The ratio in dB of an input voltage change to a "hold" mode output voltage change.
$I_{CH(+)}$, $I_{CH(-)}$ - -	Hold capacitor charge current. The current that the input amplifier can supply to charge up the hold capacitor.
$I_{HL(+)}$, $I_{HL(-)}$ - -	Hold mode leakage current. The input bias current of the output buffer amplifier. This leakage current causes a droop rate error of the external hold capacitor.
I_{IB} - - - - - - - -	Input bias current. The current flowing into the signal input for any rated common mode voltage condition.
I_{IH}, I_{IL} - - - -	Logic input current. The current into a mode control input for a forward bias (high state), I_{IH}, condition or a below threshold (low state), I_{IL}, condition.
$I_{OS(+)}$, $I_{OS(-)}$ -	Output short circuit current. The "sample" mode output short circuit current to ground with +10 V and -10 V applied at the input for $I_{OS(+)}$ and $I_{OS(-)}$, respectively.
+PSRR, -PSRR - - -	Power supply rejection ratio. The ratio in dB of the change in $+V_{CC}$ or $-V_{CC}$ voltage to the change in offset voltage measured at the output with the opposite $-V_{CC}$ or $+V_{CC}$ voltage held constant.
t_{ap} - - - - - - - -	Aperture time. The delay required between the "hold" command and a 10 volt input signal transition such that the resulting output change is less than 1 mV.
t_{aq} - - - - - - - -	Acquisition time. The time, in terms of minimum sample pulse width, that is required for the device to acquire a 10 volt full scale change to within a specified error band of final value for a specified hold capacitor size.
$TR_{(os)}$ - - - - - -	Transient response (overshoot). The percentage ratio of signal overshoot to the 100 mV final value. This parameter is related to circuit phase margin and stability.
$TR_{(ts)}$ - - - - - -	Transient response (settling time). The small signal time interval from the application of a 100 mW pulse to the time when the output enters and remains within 10 percent of its final value.
ts - - - - - - - -	Hold settling time. The time required for the output to settle within 1 mV of final value after the "hold" command is given.
V_{CM} - - - - - - - -	Common mode voltage. The voltage of the input terminal with respect to a voltage midway between $+V_{CC}$ and $-V_{CC}$.

Source: https://assist.dla.mil -- Downloaded: 20261O-21T13:28Z
Check the source to verify that this is the current version before use.

V_{HC} - - - - - - - - Hold capacitor voltage. The voltage "held" by the external hold capacitor.

V_{HS} - - - - - - - - "Hold" step voltage. The output voltage change with a fixed input voltage when the device is switched from "sample" to "hold" mode with a 4 V logic signal.

V_{IO} - - - - - - - - Input offset voltage. The "sample" mode output to input dc voltage for any rated common mode voltage condition.

6.5 Logistic support. Lead materials and finishes (see 3.3) are interchangeable. Unless otherwise specified, microcircuits acquired for Government logistic support will be acquired to device class B (see 1.2.2) and lead material and finish C (see 3.3). Longer length leads and lead forming shall not affect the part number.

6.6 Substitutability. The cross-reference information below is presented for the convenience of users. Microcircuits covered by this specification will functionally replace the listed generic-industry type. Generic-industry microcircuit types may not have equivalent operational performance characteristics across military temperature ranges or reliability factors equivalent to MIL-M-35810 device types and may have slight physical variations in relation to case size. The presence of this information shall not be deemed as permitting substitution of generic-industry types for MIL-M-35810 types or as a waiver of any of the provisions of MIL-M-38510.

Military device type	Generic-industry type
01	198
02	5537

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI
Navy - OS, SH
Air Force - 11, 19, 85, 99
DLA - ES

User activities:
Army - SM
Navy - AS, CG, MC

Agent:
DLA - ES

Preparing activity:
Air Force - 17

(Project 5962-0401)

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:28Z
Check the source to verify that this is the current version before use.

FOLD

DEFENSE LOGISTICS AGENCY

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

BUSINESS REPLY MAIL

FIRST CLASS PERMIT NO. 4966 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEFENSE LOGISTICS AGENCY

Commander
Rome Air Development Center
RADC (RBE-2)
Griffiss AFB, NY 13441

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:28Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE MIL-M-38510/125 - Microcircuits, Linear, Sample and Hold Circuits, Monolithic Silicon

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1426 1 OCT 76

EDITION OF 1 JAN 72 MAY BE USED UNTIL EXHAUSTED

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T13:28Z
Check the source to verify that this is the current version before use.